

52 Ovarian Cancer



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INCIDENCE AND ETIOLOGY

The American Cancer Society and National Cancer Institute (NCI) estimated that in 2022 there would be approximately 19,880 new cases of ovarian cancer and 12,810 women would die of this disease. The lifetime risk for epithelial ovarian cancer is 1.28%, or 1 in every 78 women. The risk is even higher among women with familial and known genetic predisposition to this disease.¹

Ovarian cancers are composed of epithelial tumors, sex cord-stromal tumors, and malignant germ cell tumors of the ovary, as well as nonspecific tumors of the ovary and metastatic cancers to the ovary.^{2,3} Nonspecific tumors, including lymphomas and sarcomas, arise from ovarian lymphocytes and fibroblasts, respectively. Among the common metastatic tumors to the ovary are cancers of the fallopian tube, endometrium, cervix, breast, appendix, and stomach or gastrointestinal tract, with the latter being known as Krukenberg tumors.³

Epithelial ovarian cancers (EOCs) are the most common ovarian cancer, accounting for approximately 90% of ovarian cancers. These malignancies arise from the coelomic epithelium that encases the ovary and share a common embryonic origin with primary carcinomas of the peritoneum.^{2,3} The most common subtype of EOC, the high-grade serous lesion, most commonly arises from the fimbriated end of the fallopian tube and is associated with mutation in breast cancer susceptibility gene 1 or gene 2 (*BRCA1* or *BRCA2*) inherited in approximately 10% to 15% of cases, with a further 5% to 10% acquired.² Although chemosensitive at diagnosis, the emergence of acquired drug resistance constitutes the greatest clinical hurdle leading to recurrent and progressive disease and, ultimately, death in the majority of patients.

Etiology

Established risk factors for EOC include age and having a family history of the disease, whereas protective factors include increasing parity, oral contraceptive pill (OCP) use, and salpingo-oophorectomy.³ Lactation, incomplete pregnancies, hysterectomy, and tubal ligation may confer a weak protective effect. Infertility may contribute to ovarian cancer risk among nulliparous women. Environmental factors may also contribute to risk because highly industrial countries have the highest reported incidences. This suggests that physical or chemical products used in industry may be causative factors. No specific environmental carcinogens that enter through the vagina or dietary factors have been identified that fulfill the requirements of causation. There is also no evidence that incriminates viruses.

Epidemiology

There is wide geographic variation in the incidence of EOC, with the highest age-adjusted incident rates (>8 per 100,000) occurring in the industrialized nations of North America and Central and Eastern Europe.⁴ Rates are lowest in Asia and Africa (≤ 3 per 100,000) and intermediate in South America (5.8 per 100,000).⁴

Interestingly, the risk increases with migration from countries with low rates to those with high rates (e.g., Japan). These observations support the theory that the causative carcinogens are in the immediate environment (e.g., food and personal customs), which may change during the cultural transition. Within the United States, rates are highest among whites, lowest among African Americans and Asians, and intermediate among Latinas.

The median age at diagnosis of EOC is between 60 and 64 years, but more than one-third of cases occur in patients aged 65 years or older.⁴ Interestingly, elderly women are more likely than younger women to be in advanced stages of ovarian cancer at initial diagnosis. Over the preceding three decades, mortality rates have decreased for women younger than age 65 years, whereas rates have increased for women older than age 65 years. This change may result from increased use of OCPs in younger patients and a shifting of the survival curve to the right.

Affected family members constitute a significant risk factor. First-degree relatives of probands have a three- to sevenfold increased risk, often with an early age at onset. The lifetime risk of EOC associated with germline *BRCA1* mutation exceeds 50% and that associated with germline *BRCA2* mutation ranges from 12% to 20%. Patients with mutations in the DNA repair genes *BRIP1*, *RAD51C*, and *RAD51D* have lifetime risks of EOC of 5.8%, 5.2%, and 12%, respectively. Approximately 2% of all cases may be associated with the hereditary nonpolyposis colorectal cancer syndrome and are commonly endometrioid or clear cell subtypes.

Hormonal and reproductive factors have been implicated in the pathogenesis of EOC, with the *incessant ovulation hypothesis* invoking a higher rate of spontaneous mutations occurring with the repair of the surface epithelium following each ovulatory cycle. Early age at menarche and late age at menopause increase the risk by increasing the total number of ovulations. Although this theory has been supported by some animal models, such as the unilateral ovulator known as the Long Island Chicken in which ovarian carcinoma is found to develop only in the ovary that ovulates, the estimated numbers of ovulatory cycles among infertile and nulliparous women does not seem to account for the full measure of ovarian carcinoma observed in the general population.³ Alternatively, the *gonadotropin hypothesis* suggests that luteinizing hormone and follicle-stimulating hormone (FSH) may be directly involved in pathogenesis. Infertility appears to be a risk factor in many epidemiologic studies, but incomplete pregnancies (i.e., spontaneous and/or induced abortions) have been associated with reduced risk in some (but not all) studies.

Because pregnancy causes anovulation and suppresses secretion of pituitary gonadotropins, both hypotheses may contain elements that account for the disease. Similarly, because lactation also suppresses secretion of pituitary gonadotropins and leads to anovulation, both hypotheses predict a protective effect from breastfeeding, which has been demonstrated in some reports. Due to their effect on ovulation, numerous epidemiologic studies have reported a consistent inverse relationship with the use of OCPs and risk of ovarian cancer. It has been estimated that OCP use has led to the prevention of 30,000 EOC cases every year and, over

the last half century, has prevented a total of 200,000 cases and 100,000 deaths. Even progestin-only contraceptives, although less well studied, have been associated with lowering the risk of ovarian cancer. Interestingly, the association of hormone replacement therapy (HRT) with ovarian cancer risk is less clear and possibly conflicting. Although HRT reduces the secretion of gonadotropins, it may also enhance estrogen-induced proliferation of ovarian cells.

Benign gynecologic conditions—including endometriosis, polycystic ovarian syndrome (PCOS), and pelvic inflammatory disease (PID)—have also been implicated in the pathogenesis of EOC. An underlying host susceptibility to implantation of exfoliated Müllerian epithelial cells from the endometrium (endometriosis) and fallopian tube (endosalpingiosis) may lead to the development of EOC, with Pearce et al.⁵ reporting an increased risk of low-grade serous ovarian cancer (odds ratio [OR], 2.11; 95% confidence interval [CI], 1.39 to 3.20), endometrioid ovarian cancer (OR, 2.04; 95% CI, 1.67 to 2.48), and clear cell ovarian cancer (OR, 3.05; 95% CI, 2.43 to 3.84). *ARID1A* mutations have been reported in endometriosis-associated ovarian carcinomas. In addition to an increased risk of endometrial cancer due to the production of unopposed endogenous estrogens and elevated androgens in women with PCOS, the Cancer and Steroid Hormone (CASH) study reported a history of PCOS among 7 of 476 (1.5%) patients with EOC compared with 24 of 4,081 controls (OR, 2.5; 95% CI, 1.1 to 5.9).⁶ Finally, although PID causes inflammation of the entire reproductive tract—including the cervix, endometrium, fallopian tubes, and ovaries—studies linking PID to EOC have been inconsistent.

Gynecologic procedures, including hysterectomy and tubal ligation, have been associated with a reduction in risk of ovarian cancer, ranging from 30% to 40%, presumably due to interruption of some of the blood flow to the ovaries. Risk-reducing salpingo-oophorectomy (RRSO) is recommended for women harboring deleterious germline *BRCA1* and *BRCA2* (*BRCA1/2*) mutations after completion of childbearing. Importantly, RRSO does not eliminate the risk entirely because there is still an approximately 1.8% to 2.0% risk for development of primary peritoneal carcinoma. Women with *BRIP1* mutations and/or strong family history with either negative *BRCA1/2* status or unknown mutation status may also benefit from RRSO.

Among environmental exposures, talcum powder has been of renewed interest recently. Talcum is a silicate, and although animal models do not support evidence of carcinogenicity, some epidemiologic meta-analyses have reported an increase in risk of approximately 35% associated with genital exposure to talc. These results were refuted by the Women's Health Initiative study in 2014 involving a cohort of 61,576 postmenopausal women, but other investigators have found a positive association with genital talc exposure in certain subgroups, including premenopausal women, postmenopausal women using HRT, and women with certain variants of glutathione S-transferase M1 and glutathione S-transferase T1.⁷

ANATOMY AND PATHOLOGY

Ovarian cancer arises in the adnexae, which consist of the ovaries, fallopian tubes, broad ligament, and embryologic rests within the broad ligament. The three main types of ovarian cancer include the epithelial cancers, malignant germ cell tumors that arise from the primordial germ cells or oocytes, and the sex cord-stromal tumors, which are derived from the steroid-producing cells responsible for nourishing the germ cells and oocytes. Interestingly, the specific malignant histologic type of ovarian cancer has less prognostic significance than the International Federation of Gynecology and Obstetrics (FIGO) stage, extent of residual disease following cytoreductive surgery, and histologic grade. Particularly in the case of EOC, histologic grade is an important independent prognostic factor. The World Health Organization (WHO) histologic classification of ovarian tumors is provided in Table 52.1.

TABLE 52.1

Previous and 2014 World Health Organization Classification of Epithelial Ovarian Tumors

Previous	New (2014)
SEROUS TUMORS	
Benign type	
Cystadenoma	Cystadenoma
Papillary cystadenoma	Adenofibroma
Surface papilloma	Surface papilloma
Adenofibroma and cystadenofibroma	
Borderline (SBOT)	
Papillary cystic BOT	SBOT/atypical proliferating tumor
Papillary surface BOT	SBOT, micropapillary type/noninvasive, serous low-grade carcinoma
Adenofibromatous and cystadenofibromatous BOT	
Malignant type	
Adenocarcinoma	Serous low-grade carcinoma
Papillary surface carcinoma	Serous high-grade carcinoma
Adenocarcinofibroma	
MUCINOUS TUMORS	
Benign type	
Cystadenoma	Cystadenoma
Adenofibroma and cystadenofibroma	Adenofibroma
Mucinous cystic tumor with mural nodules	
Mucinous cystic tumor with pseudomyxoma peritonei	
Borderline (MBOT)	
Intestinal type	MBOT/atypical proliferating mucinous tumor
Endocervical type	
Malignant type	
Adenocarcinoma	Mucinous carcinoma
Adenocarcinofibroma (malignant adenofibroma)	
ENDOMETRIOID TUMORS	
Benign type	
	Endometriosis cyst
Cystadenoma	Endometrioid cystadenoma
Adenofibroma and cystadenofibroma	Endometrioid cystadenofibroma
Borderline (EBOT)	
Cystic tumor	EBOT/atypical proliferative endometrioid tumor
Adenofibroma and cystadenofibroma	
Malignant type	
Adenocarcinoma NOS	Endometrioid carcinoma
Adenocarcinofibroma (malignant adenofibroma)	

TABLE 52.1

Previous and 2014 World Health Organization Classification of Epithelial Ovarian Tumors (continued)

Previous	New (2014)
Malignant Müllerian mixed tumor	Carcinosarcoma
Adenosarcoma	
Endometrioid stromal sarcoma (low grade)	
Undifferentiated ovarian sarcoma	
CLEAR CELL TUMORS	
Benign type	
Cystadenoma	Cystadenoma
Adenofibroma and cystadenofibroma	
Borderline (KBOT)	
Cystic tumor	CBOT/atypical proliferating clear cell tumor
Adenofibroma and cystadenofibroma	
Malignant type	
Adenocarcinoma	Clear cell tumor
Adenocarcinofibroma (malignant adenofibroma)	
Transitional cell tumors	Brenner tumors
Benign type	
Brenner tumor	Brenner tumor
Metaplastic type	
Borderline	
Borderline Brenner tumor	Borderline Brenner tumor/atypical proliferating Brenner tumor
Proliferative type	
Malignant type	
Transitional cell carcinoma	Now considered a subtype of serous cancer
Malignant Brenner tumor	Malignant Brenner tumor
SQUAMOUS EPITHELIAL TUMORS	
Mixed epithelial tumors	
Undifferentiated and unclassifiable tumors	Undifferentiated carcinoma

SBOT, serous borderline ovarian tumor; BOT, borderline ovarian tumor; MBOT, mucinous borderline ovarian tumor; EBOT, endometrioid borderline ovarian tumor; NOS, not otherwise specified; KBOT, clear cell borderline ovarian tumor; CBOT, cystic borderline ovarian tumor.

In descending order of frequency, EOCs include serous cystadenocarcinoma (characterized by psammoma bodies histologically and elevation in serum levels of the cancer antigen 125 [CA 125] clinically), endometrioid carcinoma, undifferentiated carcinoma, and clear cell carcinoma (characterized histologically by hobnail cells and coffee bean nuclei), and mucinous cystadenocarcinoma (not associated with CA 125 but may elaborate carcinoembryonic antigen [CEA]). Serous carcinomas can be high grade or low grade, and clear cell carcinomas and undifferentiated carcinomas tend to display the most aggressive behavior and confer the worst prognosis.

It is important to note that there are benign and borderline malignant counterparts of the first three tumor types. Benign lesions include serous cystadenoma, mucinous cystadenoma, and endometrioma, whereas the malignant dopplegangers include tumors of low malignant potential of each of these histologic subtypes.

SCREENING AND PREVENTION

Cancer Antigen 125

CA 125 was discovered in 1981 by Bast et al. in the 125th attempt to make a serum marker using hybridoma technology.⁸ Although it is the only U.S. Food and Drug Administration (FDA)–approved biomarker for ovarian cancer detection, it is only expressed in approximately 75% of cases and principally in the serous subtype. It is not expressed by mucinous and other ovarian carcinomas. However, when these diseases are metastatic to the abdominal cavity, levels of CA 125 may be aberrantly elevated, though not reliably enough to accurately reflect intra-abdominal tumor burden. Additional shortcomings of CA 125 include a lack of sensitivity for detecting early-stage ovarian cancer and the potential presence of this protein at abnormally high levels in many benign gynecologic and nongynecologic conditions.⁹ For these reasons, CA 125 is not a suitable screening test for ovarian cancer in the general population of women, and the search for more sensitive and informative biomarkers continues. Accepted uses of CA 125 include (1) helping to determine whether a pelvic mass is malignant; (2) assisting in determining whether a cancer of unknown primary origin has arisen from the ovary; (3) monitoring response of ovarian cancer to systemic chemotherapy; (4) carrying out surveillance of patients treated for ovarian cancer who are in remission; and (5) screening for ovarian cancer in high-risk populations (i.e., patients with a strong family history or BRCA1/2 mutation carriers).

Prostate, Lung, Colorectal, and Ovarian Cancer Screening Trial

The objective of the ovarian component of the Prostate, Lung, Colorectal, and Ovarian (PLCO) trial was to determine whether screening reduces mortality from ovarian cancer in healthy women between the ages of 55 and 74 years who still have their ovaries.⁹ A total of 34,261 women were enrolled in this trial and were randomly assigned to either no screening interventions or to yearly transvaginal ultrasounds plus CA 125. A total of 90 patients were diagnosed with ovarian cancer, of whom 60 (67%) were detected through screening with ultrasound plus CA 125. However, 72% of the screen-detected cases were late-stage ovarian cancers (i.e., stages III and IV). For every 20 women undergoing surgery, only 1 case of ovarian cancer was discovered. Unfortunately, 15% of those who underwent surgery suffered serious complications. Ultimately, it was determined that screening did not significantly reduce the mortality rate from ovarian cancer.

In 2012, Moore et al.¹⁰ studied blood samples taken from patients in the PLCO trial and reported that approximately 62% of the 65 patients who had CA 125 data available in blood samples collected less than a year before their ovarian cancer diagnosis had an elevated CA 125 level. These scientists probed the same blood samples for seven other promising biomarkers. However, even when combined with CA 125, this panel of markers was not found to be more sensitive than CA 125 alone in detecting ovarian cancer.¹⁰ In a 2017 update from the PLCO trial, Temkin et al.⁹ reported that histologic heterogeneity in this disease impacts detection and survival.

The Risk of Ovarian Cancer Algorithm

The Risk of Ovarian Cancer Algorithm (ROCA) was developed to improve the sensitivity of CA 125 in detecting EOC. A baseline CA

125 for an individual woman is used as a yardstick against which any fluctuations or changes in the CA 125 over time can be measured. Risk estimates or a ROCA score of developing ovarian cancer can then be provided by inputting these CA 125 changes into a mathematical model that includes the age of the woman. The hypothesis is that CA 125 levels should steadily increase over time in a woman who is ultimately going to develop ovarian cancer, whereas the CA 125 levels would be expected to remain typically stable or even decrease in those with noncancerous conditions. Theoretically, then, by monitoring the ROCA score carefully, the disease may be intercepted before it starts to spread, leading to higher cure rates.

The U.S. ROCA study was sponsored by the NCI's Cancer Genetics Network, the Early Detection Research Network, and the Ovarian Specialized Program on Research Excellence. In this single-arm, prospective, multicenter screening study, 4,051 women (aged 50–74 years) with no significant family history of breast or ovarian cancer underwent an annual CA 125 blood test. Based on the ROCA result, women were triaged to the next annual CA 125 test (low risk), to repeat the CA 125 test in 3 months (intermediate risk), or to a transvaginal ultrasound study with referral to a gynecologic oncologist (high risk). Based on the results of the clinical examination and the ultrasound, the gynecologic oncologist then made the decision as to whether to proceed with surgery.¹¹

The average annual rate of placement of study participants into the intermediate group was 5.8%, whereas the annual rate of referral for transvaginal ultrasonography and consultation with a gynecologic oncologist was 0.9%.¹¹ A total of 10 women underwent surgery, with 4 invasive ovarian cancers (1 with stage IA disease, 2 with stage IC disease, and 1 with stage IIB disease), 2 ovarian tumors of low malignant potential (both stage IA), 1 stage I endometrial cancer, and 3 benign ovarian tumors detected, providing a positive predictive value of 40% for detecting invasive ovarian cancer (95% CI, 12.2% to 73.8%). The specificity was 99.9% (95% CI, 99.7% to 100.0%).¹¹ All four women with invasive ovarian cancer were enrolled in the study for at least 3 years with low-risk annual CA 125 test values prior to increasing CA 125 levels.

These results are consistent with the United Kingdom Collaborate Trial of Ovarian Cancer Screening. The specificity of 99.8% reported in the United Kingdom study was similar to that of the U.S. study, as was the positive predictive value of 37.5%.¹² Although the ROCA can detect early-stage ovarian cancers in the general population, survival data are not yet mature to determine whether the ROCA can reduce mortality rates from this disease.

Eligibility criteria for the United Kingdom Familial Ovarian Cancer Screening Study (UKFOCSS) included >10% lifetime risk of ovarian cancer, age younger than 35 years, and no RRSO. In phase I, Rosenthal et al.^{13,14} showed that annual transvaginal ultrasound and CA 125 screening in women at high risk of ovarian and fallopian tube cancer lacked sensitivity for early-stage disease but may result in improved optimal debulking rates when patients are taken to surgery. For women at high risk who defer or decline RRSO, the ROCA is an acceptable alternative given its high sensitivity (94.7%) and significant stage shift (stage I to II), although its impact on mortality from ovarian cancer remains unknown.

Skates et al.¹⁵ collected data from prospective Cancer Genetics Network and Gynecologic Oncology Group (GOG) trials that screened 3,692 high-risk women (germline *BRCA1/2* mutation or strong breast or ovarian cancer family history) and evaluated serum CA 125 values drawn at 3-month intervals with the ROCA. The investigators reported that, in this high-risk population, ROCA evaluation every 3 months had better early-stage sensitivity and high specificity compared with CA 125 > 35 U/mL every 6 months to 12 months.¹⁵

Prevention of Ovarian Carcinoma

Use of OCPs has been associated with a significant reduction in the risk of ovarian cancer. Specifically, the risk has been shown to decrease by 10% to 12% after 1 year of use and by approximately

50% after 5 years of use.¹⁶ Follow-up analysis of CASH data has indicated that formulations with high levels of progestin are associated with a lower risk of ovarian cancer.⁶ The Steroid Hormones and REproductions (SHARE) study was noteworthy for finding no difference in ovarian cancer risk between androgenic and nonandrogenic pills.¹⁷ Women harboring germline *BRCA1/2* mutations also benefited from a reduction in risk of ovarian cancer through OCP use.

Risk-reducing salpingo-oophorectomy should be considered for women with germline *BRCA1/2* and *BRIP1* mutations. Women with a strong family history of either ovarian or breast cancer who have not undergone genetic testing may carry a deleterious mutation and can be presumed to be at higher-than-average risk. These women should also consider RRSO, which may also reduce the risk of breast cancer by 30% to 75%. In most situations, RRSO is typically deferred until age 35 years or when childbearing has been completed if that occurs sooner. Finally, it has been estimated that approximately 15% of patients with Lynch syndrome are at risk for ovarian cancer. These patients also have a lifetime risk of 60% for developing endometrial cancer; therefore, risk-reducing surgery includes hysterectomy.¹⁸

Due to microscopic rests of residual ovary, occult pre-existing carcinomatosis at the time of prophylactic surgery, and/or multifocal origin of peritoneal tissue after RRSO, the risk of developing serous carcinoma of the peritoneum has been reported to be in the range of 1.7% to 4.3%.¹⁸

The technique of RRSO and pathologic processing should include the following: (1) bilateral salpingo-oophorectomy with removal of the entire fallopian tube, (2) cytologic examination of peritoneal washings, (3) random peritoneal and omental biopsies along with a biopsy of any suspicious lesion, and (4) serial sectioning of the entire fallopian tube and ovaries at 2-mm intervals and microscopic examination of all sections.

Gynecologic Oncology Group protocol 0199 is a nonrandomized trial that enrolled women at high risk of developing ovarian cancer (i.e., *BRCA1/2* mutation carriers or strong family history).¹⁹ It was designed to compare RRSO at enrollment with serial transvaginal ultrasonography and CA 125 screening (ROCA). All enrolled patients had a baseline CA 125 and a transvaginal ultrasound performed and then chose to have either RRSO or continue to be screened at 3-month intervals with the ROCA evaluation. Pathologic review of the 966 prophylactic surgical specimens revealed four preinvasive tubal cancers and 20 invasive pelvic cancers involving exclusively the ovary, fallopian tube, or inner peritoneal lining of the body. Of these pelvic cancers, only 12 were detected microscopically, but all 20 of the cancers were serous carcinomas. Overall, the prevalence of serous pelvic cancers in these asymptomatic women with *BRCA1/2* mutations was 3.2% as compared with 0.5% among patients who did not have a *BRCA1/2* mutation but had strong family history of breast or ovarian cancer. Interestingly, 515 patients had their uterus removed at the time of removal of the ovaries, and 6 endometrial cancers were also found.

DIAGNOSIS

The Adnexal Mass

Although most patients with EOC will present with FIGO stage III or IV disease, occasionally, a woman with a pelvic mass in the absence of ascites, carcinomatosis, and pleural effusion or other clinical and radiologic findings of advanced disease will be ultimately diagnosed with an early-stage ovarian carcinoma. The decision to operate on a seemingly isolated pelvic mass must consider the following: patient age and symptoms, pelvic examination findings (e.g., fixed, firm, nodular mass versus a soft, smooth, mobile mass), biochemical analyses (e.g., CA 125), pelvic mass architecture on transvaginal ultrasonography, and pertinent family history concerning breast or ovarian cancer and Lynch syndrome.

The differential diagnosis of an adnexal mass includes conditions involving adjacent structures. Hydrosalpinx and paratubal cysts arising from the fallopian tube represent cystic lesions, whereas ectopic pregnancy and tubal neoplasms are examples of solid masses; tubo-ovarian abscesses can have both solid and cystic components. An intrauterine pregnancy in a bicornuate uterus may present as a cystic adnexal mass due to the fluid-filled amniotic sac, whereas a pedunculated uterine myoma may present as a solid mass in the adnexa. A distended sigmoid colon gives the appearance of a cystic mass, whereas diverticulitis or a primary colon cancer may be mistaken for a solid ovarian tumor. Finally, a distended bladder or a hydropic pelvic kidney can take on the features of a cystic adnexal mass.

Among the most common benign ovarian masses are functional cysts (e.g., corpus luteum, follicular, and theca lutein).²⁰ Endometriotic cysts, although benign, may be the source of significant pelvic pain and even infertility. Polycystic ovaries contain multiple follicle cysts with hyperplasia and luteinization of the theca interna surrounding the cysts and atretic follicles. In this condition, the ovaries may be two to five times the normal size, with a thickened capsule. PCOS may be associated with infertility and insulin-resistant diabetes mellitus.

Among the benign mixed cystic and solid ovarian neoplasms are serous cystadenomas, mucinous cystadenomas, and dermoid cysts (i.e., cystic teratoma).²⁰ Benign solid tumors of the ovary include Brenner tumor, struma ovarii, and fibroma. Meigs syndrome is characterized by ascites, hydrothorax, and an ovarian tumor (most commonly a fibroma).

Transvaginal ultrasonography is a powerful tool that allows one to determine whether an adnexal mass is solid or cystic and simple or complex; if complex, transvaginal ultrasonography can determine the complexity due to septations, excrescences, or solid components. Transvaginal ultrasonography can accurately measure the mass in two dimensions; if the mass is large enough to compress the ipsilateral ureter, hydronephrosis and/or hydroureter may also be observed. Ultrasonography can also determine if there is ascites present in the pelvis. Computed tomography (CT) imaging and/or magnetic resonance imaging (MRI) can also be used to evaluate a pelvic mass but is not always necessary, especially if the symptomatology of a patient warrants urgent operative intervention. However, many clinicians routinely order CT or MRI when there is a moderate to high suspicion for malignancy to look for signs of omental caking or carcinomatosis.²⁰ A paracentesis performed under ultrasound or CT guidance can palliate significant bloating, and cytologic analysis of the fluid may establish a diagnosis of malignant ascites. False-negative cytology in the setting of ovarian cancer may result from suboptimal laboratory protocol in preparing the cell blocks.

Conditions raising the concern of possible malignancy include bilateral disease, complex masses with sonographic evidence of solid areas, thick septations and/or mural nodules, persistent or enlarging complex masses in premenopausal patients, complex masses of any size in postmenopausal patients, masses associated with elevated tumor markers, and, finally, symptomatic masses (pelvic pain or discomfort, clear vaginal discharge, bladder, and/or gastrointestinal effects).

To assist in the evaluation and triage of a complex pelvic mass, in 2002, the American College of Obstetricians and Gynecologists (ACOG) joined with the Society of Gynecologic Oncologists (SGO) to prepare guidelines to help the general practitioner and general gynecologist direct referral when necessary to a gynecologic oncologist.²¹ Gynecologists are advised to perform a pelvic examination and imaging as appropriate for the symptoms with which the patient presents or the physical examination findings. For premenopausal women with a suspicious pelvic mass, referral to a gynecologic oncologist should be made as a result of at least one of the following: CA 125 level > 200 U/mL, ascites, abdominal or distant metastases, or one or more first-degree relatives with breast or ovarian cancer. For postmenopausal women with a concerning

pelvic mass, consultation should be considered for any CA 125 elevation, ascites, nodularity or limited mobility, evidence of metastasis, or history of a first-degree relative with breast or ovarian cancer.

Human Epididymis Protein 4

Human epididymis protein 4 (HE4) is the product of the *WFDC2* (*HE4*) gene and is overexpressed in patients with ovarian carcinoma.²² Paek et al.²³ investigated the prognostic significance of HE4 in 45 women with ovarian cancer. The elevated serum HE4 level correlated with advanced-stage disease and serous histology. The median progression-free survival (PFS) for women with elevated HE4 was 20.1 months versus 24.2 months for women with normal HE4 ($P = 0.029$).

Risk of Ovarian Malignancy Algorithm: Cancer Antigen 125 and Human Epididymis Protein 4

The Risk of Ovarian Malignancy Algorithm (ROMA) test uses two serum biomarkers to detect ovarian cancer (CA 125 and HE4). This diagnostic test should not be confused with the ROCA, discussed previously in the context of screening. In a prospective study in 472 patients with a pelvic mass, Moore et al.^{24–26} reported that 383 women were diagnosed with benign disease and 89 women were found to have cancer. This diagnostic calculator had a sensitivity of 93.8% in detecting ovarian cancer. The ROMA (CA 125 plus HE4) was FDA approved in 2011; like OVA1, it should be used for the preoperative triage of a woman with a pelvic mass and not for ovarian cancer screening. There have been no head-to-head comparisons between the two tests. Therefore, it is not known whether one or the other is superior.

OVA1 and OVA2

OVA1 was the first blood test approved by the FDA for assisting with the triage of an adnexal mass. It is a multivariate index assay that evaluates the following five biomarkers^{27,28}: CA 125 (typically increased in high-grade serous ovarian cancers), apolipoprotein A1 (decreased in ovarian tumors), β_2 -microglobulin (increased in ovarian tumors), transferrin (decreased in ovarian tumors), and prealbumin (decreased in ovarian tumors).

Important attributes of this blood test include its ability to identify different types of ovarian cancer (i.e., not only serous cancer). It also performs well in the detection of both early- and advanced-stage ovarian cancers, it may identify nonovarian cancers, and it provides a composite score that can be interpreted depending on the menopausal status of the patient on a scale of 1 to 10. An abnormal score for a premenopausal woman is >5, whereas an abnormal score for a postmenopausal woman is >4.4. The blood test was developed only for the triage of a patient with a pelvic mass who needs to undergo surgery. OVA1 was not approved to be used as a screening test.^{27,28}

The OVA1 test has been shown to outperform CA 125 testing and clinical assessment of a pelvic mass performed by a gynecologist. In a study of >500 women with a pelvic mass, Ueland et al.²⁹ reported that the combination of this blood test plus physician assessment correctly identified cancers missed by physician assessment alone and also detected 76% of cancers missed by CA 125 testing alone. Next, Bristow et al.³⁰ conducted a prospective, multi-institutional trial designed to validate the effectiveness of OVA1 in identifying ovarian malignancy compared with clinical assessment and CA 125 testing among women undergoing surgery for an adnexal mass. In this study, which included 494 patients, the study team reported that OVA1 demonstrated higher sensitivity (95.7%) and negative predictive value (98.1%) than clinical impression and CA 125 testing.³⁰

Most recently, the algorithm was optimized to increase specificity by substituting FSH and HE4 for β_2 -microglobulin and

prealbumin. CA 125 II, apolipoprotein A1, and transferrin remain in this second-generation test named OVA2 or Overa. OVA2 uses a single cutoff point of 5 and does not require physicians to determine the menopausal status of the patient. OVA2 was cleared by the FDA on March 21, 2016.

PRESENTATION AND EVALUATION OF ADVANCED DISEASE

The most common presenting symptoms of ovarian cancer in decreasing order of frequency are abdominal swelling, abdominal pain, dyspepsia, urinary frequency, and weight change. Unfortunately, the presence of any of these symptoms often is indicative of advanced disease. Although there is no reliable constellation of symptoms indicative of early disease, patient advocacy efforts urge women to “listen for the whisper.” Certainly, a high index of suspicion on the part of the physician can sometimes lead to early intervention in any woman between the ages of 40 and 70 years who has persistent gastrointestinal symptoms. The problem, of course, is that most nonspecific abdominal and pelvic complaints are not malignant, and it is not cost-effective to order CT scans on every woman with a presumptive diagnosis of irritable bowel syndrome.

In addition to abdominal bloating resulting from malignant ascites, many women with ovarian carcinoma will experience significant gastrointestinal disturbance and may present with signs and symptoms of partial small bowel obstruction. Others may present with unexplained or unintended weight loss due to decreased oral intake as a result of intra-abdominal pressure on the stomach and intestinal tract and obstructive symptoms making it hard to keep meals down. Loss of appetite is common in these cases. Finally, some patients will present with shortness of breath due to a malignant pleural effusion, whereas others may have the diagnosis of carcinoma made simultaneously with the occurrence of a thromboembolic event. The eminent German physician Rudolf Virchow (1821–1902) proposed the triad of hypercoagulability, stasis or turbulence of blood flow, and endothelial injury—all of which are associated with malignancy—to elucidate the etiology of pulmonary embolism.³¹

The decision to explore a patient with carcinomatosis likely to have arisen from the adnexae can be made only by a gynecologic oncologist. A methodical assessment of the preoperative nutritional and performance status of the patient is imperative. Preoperative serum albumin and prealbumin should be obtained, along with a complete blood count with differential, comprehensive metabolic panel, coagulation panel, liver function tests, blood type and screen for antibodies, a CA 125 test, and urinalysis with culture. A multidisciplinary collaboration is incumbent throughout preoperative medical optimization, with the gynecologic oncologist working directly with the patient's primary care physician and other subspecialist physicians (e.g., cardiologist) as needed. Personalized review of all films with the radiologist can be helpful. Multiple parenchymal liver metastases, extensive suprarenal adenopathy, porta hepatis disease, bony metastases, pulmonary masses, and any other extra-abdominal metastases are relative contraindications to primary cytoreduction. Such patients should be treated with neoadjuvant chemotherapy followed by interval cytoreduction if there is radiologic, biochemical, and/or physical evidence of objective response. Once a decision to proceed with surgery has been made, however, it is necessary to first correct malnutrition, anemia, coagulopathy, and electrolyte aberrations. In addition, urinary tract infection should be treated if present. Preoperative thoracentesis to address a pleural effusion and inferior vena cava filter placement for any patient who presents with thrombosis may also be required.

When the clinical picture is unclear, combined positron emission tomography (PET) and CT or even diagnostic laparoscopy may be used to determine whether it is possible to surgically resect all gross disease while minimizing perioperative morbidity. Because of the risks of inadvertent trocar injury to the bowel and

tumor seeding along the port track, laparoscopy should be considered in the setting of widespread carcinomatosis and large-volume malignant ascites as core biopsy and neoadjuvant chemotherapy is popular, though many clinicians believe that every patient who does not clearly have stage IV disease deserves laparoscopic evaluation of resectability. Occasionally, intra-abdominal tuberculosis, yersinia or pseudomyxoma peritonei from a ruptured appendiceal neoplasm will be included in the differential diagnosis.

INTERNATIONAL FEDERATION OF GYNECOLOGY AND OBSTETRICS STAGING

FIGO issued a revised surgical staging system for EOC, which was released on January 1, 2014 (Table 52.2).³² Because there are no reliable early symptoms and because no validated screening tests are available for the general population, most women with EOC

TABLE 52.2

International Federation of Gynecology and Obstetrics (FIGO) Staging of Ovarian, Fallopian Tube, and Primary Peritoneal Cancer^a

FIGO (1988)	FIGO (2014)
I: Tumor confined to the ovaries	I: Tumor confined to the ovaries or tube(s)
IA: Tumor confined to 1 ovary (capsule intact), no tumor on ovarian surface, no malignant cells in the ascites or peritoneal cytology	IA: Tumor confined to 1 ovary or tube (capsule intact), no tumor on ovarian surface, no malignant cells in the ascites or peritoneal cytology
IB: Tumor involves both ovaries (capsule intact), no tumor on ovarian surface, no malignant cells in the ascites or peritoneal cytology	IB: Tumor involves both ovaries or tubes (capsule intact), no tumor on the ovarian or tubal surface, no malignant cells in the ascites or peritoneal cytology
IC: Tumor limited to 1 or both ovaries together with 1 of the following: capsule rupture; tumor on ovarian surface; malignant cells in the ascites or peritoneal cytology	IC: Tumor limited to 1 or both ovaries or tube(s) together with 1 of the following: ■ IC1: Capsule rupture intraoperatively ■ IC2: Capsule rupture preoperatively or tumor on ovarian or tubal surface ■ IC3: Malignant cells in the ascites or peritoneal cytology
II: Tumor in 1 or both ovaries with pelvic involvement	II: Tumor in 1 or both ovaries or tube(s) with pelvic involvement or primary peritoneal carcinoma
IIA: Extension and/or implant on uterus and/or fallopian tube(s); no malignant cells in the ascites or peritoneal cytology	IIA: Extension and/or implant on uterus and/or fallopian tube(s) and/or ovary(ies)
IIB: Extension to other pelvic tissue; no malignant cells in the ascites or peritoneal cytology	IIB: Extension to other pelvic intra peritoneal tissue
IIC: IIA or IIB plus malignant cells in the ascites or peritoneal cytology	

TABLE 52.2

International Federation of Gynecology and Obstetrics (FIGO) Staging of Ovarian, Fallopian Tube, and Primary Peritoneal Cancer^a (continued)

FIGO (1988)	FIGO (2014)
III: Tumor in 1 or both ovaries with microscopic extrapelvic peritoneal metastases and/or regional lymph node metastases	III: Tumor in 1 or both ovaries or tube(s) or primary peritoneal carcinoma with cytologically or histologically verified peritoneal metastases outside the pelvis and/or retroperitoneal lymph node metastases
IIIA: Microscopic extrapelvic peritoneal metastases	IIIA1: Positive retroperitoneal lymph nodes only (verified cytologically or histologically) ■ IIIA1(i): Metastases, maximum diameter 10 mm ■ IIIA1(ii): Metastases, maximum diameter >10 mm IIIA2: Microscopic extrapelvic peritoneal metastases with or without positive retroperitoneal lymph nodes
IIIB: Macroscopic extrapelvic peritoneal metastases of up to 2 cm	IIIB: Macroscopic peritoneal pelvic metastases of up to 2 cm, with or without retroperitoneal lymph node metastases (including the capsule of the liver/spleen but excluding parenchymatous metastasis)
IIIC: Macroscopic extrapelvic peritoneal metastases larger than 2 cm and/or regional lymph node metastases	IIIC: Macroscopic extrapelvic peritoneal metastases larger than 2 cm with or without retroperitoneal lymph node metastases (including capsule of the liver/spleen but excluding parenchymatous metastases)
IV: Distant metastases excluding peritoneal metastases	IV: Distant metastasis excluding peritoneal metastases IVA: Pleural effusion with positive cytology IVB: Parenchymatous metastases and metastases to extra-abdominal organs (including inguinal lymph node metastases and extra-abdominal lymph node metastases)

^aFIGO 2014 versus FIGO 1988; differences appear in bold.

present with advanced-stage disease, usually FIGO stage IIIC (i.e., abdominal spread with lesions >2 cm in maximal diameter). The situation is usually even more discouraging because the metastatic deposits are occasionally very large and adherent to multiple organs and structures throughout the abdomen and pelvis. The omentum is usually extensively involved such that at laparotomy the presentation may at first appear hopeless. Primary cytoreductive surgery

should be carried out if preoperative imaging and physical examination suggest that there is a high probability of complete resection (i.e., R₀ or no gross residual).

In the uncommon event that a seemingly isolated pelvic mass suspicious for ovarian carcinoma is discovered, it is the obligation of the surgeon to remove it, ensuring that it remains intact. If intraoperative rapid mandatory frozen section analysis is positive for carcinoma, then surgical staging is imperative. Up to 30% of clinical stage I ovarian carcinomas will be upstaged based on comprehensive surgical staging, which includes total abdominal hysterectomy, bilateral salpingo-oophorectomy, total omentectomy, bilateral pelvic and para-aortic lymphadenectomy, peritoneal biopsies, and four-quadrant pelvic washings.³³ Interestingly, many cases of upstaging are due to occult para-aortic lymph node metastases identified by the pathologist after surgical staging.

MANAGEMENT BY STAGE

Early-Stage Epithelial Ovarian Cancer (FIGO Stage I)

Unfortunately, only 10% of patients with EOC are diagnosed with early-stage disease (i.e., FIGO stage I). For women with non—clear-cell, FIGO stage IA or IB, grade 1 or 2 malignancies, provided they have undergone comprehensive surgical staging, adjuvant chemotherapy is not required. For all other early-stage patients (i.e., FIGO stage IC or grade 3 or any clear cell) as well as patients who have not been staged surgically, chemotherapy is indicated.

GOG protocol 157 was designed to study the number of cycles of adjuvant platinum-paclitaxel chemotherapy in women with early-stage ovarian carcinoma who had undergone comprehensive surgical staging to determine the impact on rate of recurrence and to compare toxicities.³⁴ At a median follow-up of 6.8 years, the recurrence rate for six cycles of adjuvant chemotherapy was 24% lower (hazard ratio [HR], 0.761; 95% CI, 0.51 to 1.13; *P* = 0.18) than for three cycles, and the overall death rate was similar in the two arms (HR, 1.02; 95% CI, 0.662 to 1.57).³⁴ Grade 3 or 4 neurotoxicity occurred in 4 of 211 (2%) and 24 of 212 (11%) patients treated on the 3- and 6-cycle regimens, respectively (*P* < 0.01). Six cycles also caused significantly more severe anemia and granulocytopenia.⁷ For surgically staged high-risk, early-stage ovarian carcinoma, this trial was practice changing because it demonstrated that six cycles of adjuvant chemotherapy were associated with higher toxicity and did not significantly alter the recurrence rate in this population. However, in an analysis of patients with high-grade serous tumors, the 5-year recurrence-free survival was 83% and 60% in those who received 6 versus 3 cycles of chemotherapy, respectively (*P* = 0.007).³⁵ For that reason, six cycles are recommended for high-grade serous tumors.

National consensus guidelines are available on a variety of websites. In 2013, Bristow et al.³⁶ performed a Surveillance, Epidemiology, and End Results (SEER) study in California and found that most women with ovarian cancer undergoing surgery in this state were being operated on by surgeons who were not trained to perform the type of operation required to adequately stage patients and completely remove all visible disease when it was found to have spread throughout the abdominal cavity.³⁶ This is unfortunate because only one-third of patients studied were referred to and treated by gynecologic oncologists who were considered to be high-volume surgeons. High-volume surgeons were characterized as having treated over 10 to 15 cases per year in institutions treating more than 20 cases per year.³⁶ The study indicated that patients with stages I and II disease who were not treated according to the consensus guidelines had worse survival rates compared with those who were treated according to the guidelines. Survival was also worse for patients with advanced-stage cancers.³⁶

Uncommon and Rare Tumors of the Ovary

Also called *borderline tumors*, tumors of low malignant potential can be of serous, mucinous, or endometrioid histology and are commonly found in reproductive-age and middle-aged women. For women who have not completed childbearing, fertility-preserving surgery can be offered. Otherwise, bilateral adnexectomy is curative in this disease. Advanced disease may manifest with either noninvasive or invasive peritoneal and omental implants. Because these tumors have a small population of cells in the actively dividing pool, they tend to be resistant to most antineoplastic agents that are cell-cycle specific. As a result, late recurrence is common in this tumor class. The Arbeitsgemeinschaft Gynaekologische Onkologie (AGO) Retrospective/Prospective Multicenter Outcome Survey in Borderline Ovarian Tumors (ROBOT) study is the largest data set available on borderline ovarian tumors ($n = 950$ pathologically confirmed on review). Recurrence occurred in 7.8% of patients ($n = 74$), with disease progression in the form of invasive carcinoma occurring in 30% of relapses.²⁰ Inadequate surgical staging, residual tumor, fertility-sparing surgery, and high FIGO stage were associated with shorter PFS. No differences were observed for laparotomy versus laparoscopy or application of adjuvant chemotherapy. Mortality was 4.5% ($n = 43$).²⁰

Malignant germ cell tumors of the ovary account for approximately 5% of all ovarian malignancies, with a median age of presentation between 19 and 21 years.²⁰ They are almost always unilateral and are exquisitely chemosensitive. For this reason, fertility-preserving surgery is always an option for this disease, even when metastases are present. Among the different subtypes are the radiosensitive dysgerminoma (bilateral in 10% of cases, elaborates lactate dehydrogenase and alkaline phosphatase, and is characterized by lymphocytic infiltration), the yolk sac tumor (spreads hematogenously, elaborates α -fetoprotein, and is characterized histologically by Schiller-Duval bodies), the immature teratoma (forms neuroepithelial rosettes that serve as the basis for the Norris grading system and elaborates both α -fetoprotein and human chorionic gonadotropin), the mixed germ cell tumor, embryonal carcinoma, nongestational choriocarcinoma, and the dreaded polyembryoma. Therapy for malignant germ cell tumors mirrors and has been extrapolated from the more common testicular germ cell cancer experience. Except for FIGO surgical stage IA dysgerminoma and stage IA, grade 1 or 2 immature teratoma, patients with malignant germ cell tumors of the ovary should be treated with adjuvant bleomycin-etoposide-cisplatin. Prognosis is excellent for patients with this class of ovarian cancer, with return of normal menstrual function being the rule and numerous successful pregnancies documented during the follow-up of the survivors.

Accounting for less than 5% of ovarian malignant neoplasms, sex cord-stromal tumors can be broadly separated into the estrogen-secreting tumors (granulosa cell tumor [GCT] and juvenile granulosa cell tumor [JGCT]) and the testosterone-secreting tumors (Sertoli-Leydig tumors).²⁰ Although GCTs are usually diagnosed in women in their mid-50s, whereas JGCTs typically occur in children and adolescents, both types of tumor have been reported across the age spectrum. Histologically, coffee bean nuclei and Call-Exner bodies are observed. Serum levels of inhibin A (and occasionally inhibin B) can assist in the preoperative diagnosis of GCTs and can also be used to monitor patients for recurrence. Occasionally, the estrogen made by a GCT can induce neoplastic changes in the endometrium and manifest as uterine bleeding from hormonally driven atypical endometrial hyperplasia or carcinoma. In children, hormone production by a JGCT may result in isosexual precocious puberty. Lymphadenectomy can probably be omitted from surgical staging of these tumors in the absence of grossly enlarged nodes. Most GCTs contain a mutation in the *FOXL2* gene. The presence of the GSP oncogene and high GATA binding protein 4 (GATA) expression in JGCTs may provide new prognostic markers. Among young patients with JGCTs,

fertility-preserving surgery should be instituted.²⁰ Although most patients with GCTs present with early-stage disease and can be cured, over 80% of patients with advanced disease will die. The bleomycin, etoposide, and cisplatin (BEP) regimen used for malignant germ cell tumors of the ovary has demonstrable activity in GCTs, but the responses have not been durable and long lasting. Sertoli-Leydig cell tumors can cause virilization due to high levels of circulating testosterone. Surgical resection is usually curative. Patients with advanced Sertoli-Leydig cell tumors may respond to gonadotropin-releasing hormone analogs.²⁰ Gynandroblastoma is an exceedingly rare primary sex cord-stromal tumor of the ovary showing morphologic evidence of both male and female differentiation. Finally, gonadoblastoma is another rare tumor that has both primordial germ cells and immature Sertoli or granulosa cells.

Advanced-Stage Epithelial Ovarian Cancer (FIGO Stages II to IV)

The historic treatment paradigm for women diagnosed with advanced EOC is a model in evolution. Recently published data from the SEER Program of the NCI indicate that only 15% of women are diagnosed with localized disease (FIGO stage I), for which the 5-year relative survival rate approaches 92.3%.³⁷ Among the 18% of patients with regional spread of the disease to adjacent pelvic organs (FIGO stage II), 5-year relative survival is 71.7%. For 61% of patients, the disease has spread to distant organs in the abdomen and/or lymph nodes (FIGO stage III), and the 5-year relative survival rate is only 27.4%.³⁷ Investigators at the Centers for Disease Control and Prevention have reported on mortality trends resulting from ovarian cancer from 2000 to 2009, noting that deaths due to this disease have decreased significantly, by 1.5% per year, among women overall and among non-Hispanic white women. Mortality rates among Hispanic women were found to have decreased significantly, by 1.1% per year.³⁸ The mortality rates remained stable among African American, American Indian/Alaskan Native, and Asian/Pacific Islander women.³⁷

Patients often present with a constellation of symptoms consistent with advanced disease. Most patients undergo radiographic imaging, most commonly with pelvic ultrasonography and/or CT imaging. A definitive diagnosis is often made at the time of surgical exploration, but cytology from abdominal paracentesis and/or histologic confirmation of disease through CT-guided biopsy of a metastatic disease site can also be informative.

Cytoreductive surgery and chemotherapy are required in the majority of cases of advanced-stage disease. The traditional treatment paradigm is to operate prior to administration of six to eight cycles of adjuvant platinum- and taxane-based systemic therapy, but this approach has been revisited during the preceding decade as the definition of what constitutes optimal surgical cytoreduction has evolved along with the feasibility of neoadjuvant chemotherapy. An important area of study involves mapping the mutational landscape of the histologic subtypes of this disease (Table 52.3). Although at least 75% of women with advanced disease can enter remission after surgery and chemotherapy, due to acquired drug resistance, the majority of patients will relapse and ultimately die from disease. Ten-year disease-free survival ranges from 5% to 10%.

Significant strides have been made with incorporation of targeted therapy in the management algorithms for advanced ovarian carcinoma, with FDA approvals and regulatory approval abroad having been granted for antiangiogenesis therapy and integration of poly (ADP-ribose) polymerase I (PARP) inhibitors (PARPi). Randomized clinical trials in the field of immuno-oncology using checkpoint inhibitors are also ongoing. Although these targeted agents have been studied as part of primary therapy and recurrent disease, a notable emerging role may actually be as maintenance therapies after complete response (CR) or partial response (PR) to standard chemotherapy. For this reason, the targeted agents are discussed separately after a detailed review of the chemotherapeutic

TABLE 52.3

Subtypes of Ovarian Carcinoma and Associated Genetic Changes

Histologic Subtype	High-Grade Serous	Low-Grade Serous	Endometrioid	Clear Cell	Mucinous
Mutation(s)	TP53, BRCA1/2, RAD51C, RAD51D, BARD1, PALB2	KRAS/BRAF, ERBB2, PIK3CA	ARID1A, CTNNB1, PTEN, PIK3CA, PPP2RIA	ARID1A, PIK3CA, ZNF217, PPP2RIA	KRAS

agents used in this disease and the clinical trials in which efficacy and toxicology were validated.

MANAGEMENT OF NEWLY DIAGNOSED ADVANCED-STAGE DISEASE

Cytoreductive Surgery

Although the term *optimal cytoreduction* has been used throughout the years to represent the goal of surgery to leave behind ≤ 1 to 2 cm of tumor volume, the modern school of thought mandates that the goal of cytoreductive surgery is to leave behind no gross residual disease (i.e., R_0 or complete resection). Interestingly, the first evidence that cytoreductive surgery impacts survival can be found in the 1973 report by Magrath et al.³⁹ on the survival advantage conferred by extensive disease resections in pediatric patients in Africa with abdominal Burkitt lymphoma. This was followed by the landmark study by Griffiths⁴⁰ from 1974 in which the surgeon reported that among 102 consecutive cases of advanced ovarian cancer, patients in whom residual disease greater than 1.5 cm in maximal diameter was left in the abdominal cavity had all died from disease within 2 years compared with the 20% 2-year survival rate afforded by residual tumor burden less than 1.5 cm.⁴⁰ The watershed event occurred in 2002 when Bristow et al.⁴¹ performed a meta-analysis of 81 ovarian cancer cohorts (nearly 7,000 patients) treated with platinum-based chemotherapy and demonstrated that the most important determinant of survival was maximal cytoreduction, with each 10% increase in tumor resection associated with a 5.5% increase in median survival (Fig. 52.1).⁴¹

Primary debulking may require many hours of operative time and be associated with multivisceral resections, packed red cell transfusions, admission to the intensive care unit with cardiac

monitoring, and even, in the most experienced hands, significant postoperative sequelae. However, the outcome of the surgical approach in terms of residual disease remains the most important prognostic factor for survival. In 2017, Harter et al.⁴² reported on the Lymphadenectomy in Ovarian Neoplasms (LION) study. Patients who underwent optimal cytoreduction and had clinically negative lymph nodes were randomized intraoperatively to undergo formal lymphadenectomy versus no lymphadenectomy. The study's primary end point of overall survival (OS) was not met (median, 66 months in the lymphadenectomy arm and 69 months in the no-lymphadenectomy arm; HR, 1.06; 95% CI, 0.83 to 1.34; $P = 0.65$),⁴² but postoperative morbidity and mortality were significantly higher in the lymphadenectomy arm, indicating that systemic lymphadenectomy of clinically negative lymph nodes in women who have undergone complete resection should be omitted.

For primary disease, surgery can be performed upfront or after three to four cycles of neoadjuvant chemotherapy. For patients for whom remission and/or cure is highly unlikely, surgery can be palliative (e.g., small bowel resection or ileostomy for obstruction or gastric tube placement).

In the United States, most gynecologic oncologists perform surgery prior to chemotherapy and use neoadjuvant chemotherapy followed by interval cytoreduction and postoperative chemotherapy for those who present with what seems to be unresectable disease and for patients whose debilitated medical condition precludes their candidacy for upfront cytoreduction.

Patients in remission who ultimately experience recurrence may also be candidates for additional cytoreductive surgery or secondary debulking. Ideally, such patients should have had a prolonged disease-free interval (e.g., a remission lasting ≥ 12 months), have small-volume or isolated recurrent disease as opposed to widespread carcinomatosis, and be medically optimized to undergo secondary debulking. Following these same general guidelines, tertiary cytoreductions may also be possible.

Irrespective of where surgery is positioned in the disease course, all visible tumor should be removed.

LION (NCT00712218)

LION is a randomized trial to evaluate systematic pelvic and para-aortic lymphadenectomy in 647 women with FIGO stage IIB through IV ovarian cancer who had undergone macroscopic complete resection and had normal appearing lymph nodes both preoperatively and intraoperatively. Randomization to lymphadenectomy or no lymphadenectomy occurred intraoperatively following complete surgical cytoreduction. The median PFS was 25.5 months in both groups (HR, 1.11; 95% CI, 0.92 to 1.34; $P = 0.29$) and median OS (primary endpoint) was 69.2 months in the no-lymphadenectomy group and 65.5 months in the lymphadenectomy group (HR, 1.06; 95% CI, 0.83 to 1.34; $P = 0.65$). Importantly, serious postoperative complications were higher in the lymphadenectomy cohort: repeat laparotomy, 12.4% versus 6.5% ($P = 0.01$); and mortality within 60 days after surgery, 3.1% versus 0.9% ($P = 0.049$). As a result of LION, we recommend assessing lymph nodes using preoperative imaging modalities together with intraoperative inspection and palpation. If lymph nodes are not enlarged among patients with advanced disease who will require adjuvant therapy, they should not be systematically removed following complete resection of all visible disease.

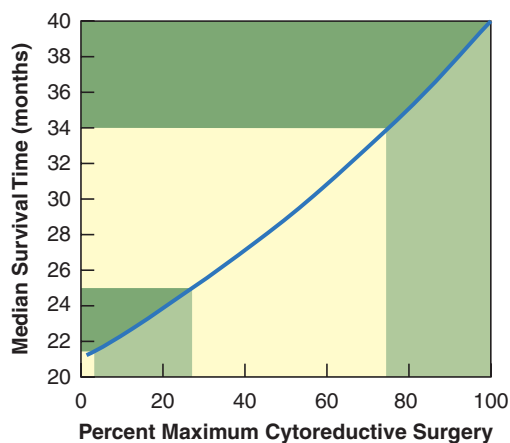


Figure 52.1 Linear regression model demonstrating that each 10% increase in maximal cytoreduction is associated with a 5.5% increase in median survival time. (Reprinted with permission. © American Society of Clinical Oncology, 2002. All rights reserved. Bristow RE, Tomacruz RS, Armstrong DK, et al. Survival effect of maximal cytoreductive surgery for advanced ovarian carcinoma during the platinum era: a meta-analysis. *J Clin Oncol* 2002;20[5]:1248–1259.)

Platinum-Taxane–Based Chemotherapy

Table 52.4 lists systemic chemotherapy (and targeted therapy) regimens used in the management of newly diagnosed and recurrent ovarian cancer. The use of platinum-taxane–based chemotherapy is the gold standard for adjuvant treatment of advanced-stage EOC. Although these drugs are used for all advanced-stage EOCs, they are probably most active in high-grade serous and undifferentiated adenocarcinomas. In 1996, McGuire et al.⁴³ reported on GOG protocol 0111, a phase III randomized trial comparing

cisplatin (75 mg/m²) with either cyclophosphamide (750 mg/m²) or paclitaxel (135 mg/m² over 24 hours) in patients with advanced disease with residual disease larger than 1 cm after surgery. Among 386 patients treated in the study, in the cisplatin-paclitaxel arm versus the cisplatin-cyclophosphamide arm, there were significant improvements in both PFS (median, 18 vs. 13 months, respectively; $P < 0.001$) and OS (median, 38 vs. 24 months, respectively; $P < 0.001$).

GOG protocol 158 was a phase III noninferiority trial designed to test the efficacy and tolerability of substitution of carboplatin

TABLE 52.4

Systemic Therapy Regimens Used in Ovarian Cancer

Newly diagnosed ovarian cancer	Bleomycin 30 U IV weekly + etoposide 100 mg/m ² days 1–5 + cisplatin 20 mg/m ² days 1–5 (malignant germ cell tumors of the ovary)
	Carboplatin AUC 4–6 + paclitaxel 175 mg/m ² 3-h infusion every 21 d
	Carboplatin AUC 5 + docetaxel 75 mg/m ² 1-h infusion every 21 d
	Carboplatin AUC 5 + paclitaxel 175 mg/m ² + bevacizumab 15 mg/kg every 21 d followed by maintenance bevacizumab 15 mg/kg every 21 d (U.S. as of June 13, 2018)
	Carboplatin AUC 6 + paclitaxel 175 mg/m ² + bevacizumab 7.5 mg/kg every 21 d followed by maintenance bevacizumab 7.5 mg/kg every 21 d (Europe only: suboptimal FIGO III and FIGO IV)
	Carboplatin AUC 6 every 21 d + paclitaxel 80 mg/m ² weekly (dose-dense paclitaxel)
	Cisplatin 75–100 mg/m ² 41°C for 30 min to 1 h IP (heated IP chemotherapy)
	Oxaliplatin (85 mg/m ²) day 1 + L-folinic acid (100 mg/m ²) followed by bolus 5-FU (400 mg/m ²) + 5-FU (600 mg/m ²) 22-h continuous infusion for 2 consecutive days (pseudomyxoma peritonei)
	Paclitaxel 135 mg/m ² 24-h infusion IV day 1 + CDDP 100 mg/m ² IP day 2 + paclitaxel 60 mg/m ² IP day 8 every 21 d (IP + IV therapy)
Platinum-resistant recurrence	Gemcitabine 800–1,000 mg/m ² days 1 and 8 every 21 d
	Nanoparticle albumin-bound paclitaxel 100 mg/m ² days 1, 8, and 15 every 28 d
	Olaparib 300 mg PO BID
	Oxaliplatin 85 mg/m ² day 1 + leucovorin 200 mg/m ² day 1 followed by bolus 5-FU (400 mg/m ²) day 1 and 5-FU (600 mg/m ²) 22-h continuous infusion (mucinous ovarian)
	Paclitaxel 80 mg/m ² days 1, 8, 15, and 22 every 4 wk + bevacizumab 10 mg/kg every 2 wk
	Paclitaxel 80 mg/m ² weekly
	Pemetrexed 500–900 mg/m ² every 21 d
	PLD 40 mg/m ²
	PLD 30 mg/m ² + trabectedin 1.1 mg/m ² every 21 d (Europe only)
	PLD 40 mg/m ² every 4 wk + bevacizumab 10 mg/kg every 2 wk
	Rucaparib 600 mg PO BID
	Topotecan 4 mg/m ² days 1, 8, and 15 every 4 wk + bevacizumab 10 mg/kg every 2 wk
Platinum-sensitive recurrence	Topotecan 4 mg/m ² weekly
	Carboplatin AUC 4 day 1 + gemcitabine 1,000 mg/m ² days 1 and 8 every 21 d
	Carboplatin AUC 4 + gemcitabine 1,000 mg/m ² + bevacizumab 15 mg/kg every 21 d followed by maintenance bevacizumab 15 mg/kg every 21 d
	Carboplatin AUC 5 + paclitaxel 175 mg/m ² + bevacizumab 15 mg/kg every 21 d followed by maintenance bevacizumab (15 mg/kg) every 21 d
	Carboplatin AUC 5 + PLD 30 mg/m ² every 28 d
	Carboplatin AUC 4–5 + paclitaxel 175 mg/m ² 3-h infusion every 21 d
	CDDP 30 mg/m ² days 1 and 8 + gemcitabine 600 mg/m ² days 1 and 8
	CDDP 75 mg/m ² + paclitaxel 135 mg/m ² 24-h infusion every 21 d
Platinum-sensitive recurrence oral maintenance therapy	Niraparib 200–300 mg once daily
	Olaparib 300 mg PO BID
	Rucaparib 600 mg PO BID

IV, Intravenous; AUC, area under the curve; FIGO, International Federation of Gynecology and Obstetrics; IP, intraperitoneal; 5-FU, 5-fluorouracil; CDDP, cisplatin; PO, oral; BID, twice a day; PLD, pegylated liposomal doxorubicin.

for cisplatin in combination with paclitaxel for advanced EOC. Gastrointestinal, renal, and metabolic toxicity and grade 4 leukopenia were more frequent in the cisplatin-containing arm, whereas grade 2 or greater thrombocytopenia was more common in the carboplatin-containing arm. Neurologic toxicity was similar in both regimens. Because carboplatin plus paclitaxel resulted in less toxicity, was easier to administer, and was not inferior to cisplatin plus paclitaxel, the combination of carboplatin plus paclitaxel emerged as the standard of care for advanced ovarian carcinoma.⁴⁴ As discussed earlier, this combination is also used for adjuvant therapy of high-risk early-stage disease (i.e., FIGO stage IA or IB grade 3, stage IC any grade, all clear cell carcinoma, and completely resected stage II). These combinations are given on a 21-day schedule for a total of six to eight cycles.

The Scottish Randomized Trial in Ovarian Cancer (SCOTROC) was a phase III, randomized, noninferiority trial to evaluate the efficacy and tolerability of substitution of docetaxel for paclitaxel when combined with carboplatin for FIGO stages IC to IV ovarian carcinoma. At a median follow-up of 23 months, both groups had similar PFS, OS, and objective tumor or CA 125 response rates. The docetaxel-carboplatin arm was associated with substantially less overall and grade 2 or higher neurotoxicity than paclitaxel-carboplatin (neurosensory, 11% vs. 30%, $P < 0.001$; neuromotor, 3% vs. 7%, $P < 0.001$).⁴⁵ Docetaxel-carboplatin was associated with significantly more grade 3 or 4 neutropenia and neutropenic complications. Based on these results, the combination of docetaxel-carboplatin has become an accepted alternative frontline regimen, especially for patients with pre-existing neuropathy (e.g., diabetic patients).

Sequential Doublets and New Triplets

Given the demonstrable activity of newer cytotoxic agents in recurrent disease, Bookman et al.⁴⁶ reported on the efficacy of adding a third agent to primary therapy in GOG protocol 182. Unfortunately, there were no improvements in either PFS or OS associated with any experimental regimen.⁴⁶ Survival analyses of groups defined by size of residual disease also failed to show experimental benefit in any subgroup.

Neoadjuvant Chemotherapy

Patients in whom malnutrition is significant or the metastatic disease process appears unresectable may be treated with neoadjuvant chemotherapy followed by interval cytoreduction and postoperative chemotherapy. However, there has been significant movement in considering neoadjuvant chemotherapy for women who are candidates for even primary debulking surgery.

A European Organisation for Research and Treatment of Cancer (EORTC) prospective randomized trial of 632 patients reported by Vergote et al.⁴⁷ demonstrated that neoadjuvant chemotherapy followed by interval debulking surgery was not inferior to the standard procedure. The HR for death in the group assigned to neoadjuvant chemotherapy followed by interval debulking was 0.98 (90% CI, 0.84 to 1.13; $P = 0.01$ for noninferiority), and the HR for PFS was 1.01 (90% CI, 0.89 to 1.15).⁴⁷ Although postoperative rates of adverse effects and mortality tended to be higher after primary debulking than after interval debulking, this study raised several controversies, particularly regarding the quality of debulking surgery. Importantly, complete resection of all macroscopic disease (at primary or interval surgery) was the strongest independent variable in predicting OS. However, only 41.6% of patients were rendered optimally debulked to 1 cm or less of residual tumor after primary cytoreduction as compared with 80.6% of patients after interval cytoreduction. Based on these results, neoadjuvant chemotherapy may be a reasonable option for significantly malnourished and other patients, including those not medically

optimized, as well as women with unresectable abdominal disease or extra-abdominal metastases.

The randomized Medical Research Council Chemotherapy or Upfront Surgery (MRC CHORUS) noninferiority trial showed no significant differences in either OS or PFS, and the authors concluded that neoadjuvant chemotherapy was noninferior to primary cytoreduction.⁴⁸ The results should be interpreted with caution given the low microscopic residual disease rate in the primary surgery arm and the identical median surgical times in both arms of 120 minutes.

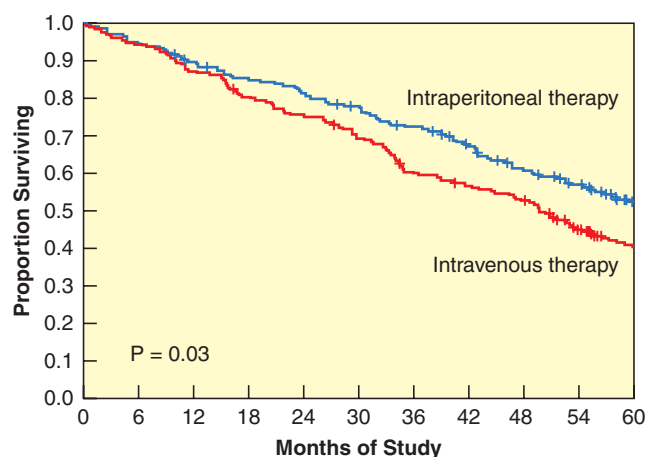
When the combination of carboplatin-paclitaxel or carboplatin-docetaxel is used for neoadjuvant chemotherapy, typically three to four cycles are given, and if there is clinical and radiologic evidence of response, an interval debulking operation is performed after 4 weeks to allow for bone marrow recovery. Two to 3 weeks after surgery, patients can resume systemic therapy with an additional three to four cycles of chemotherapy.

Intraperitoneal Chemotherapy

Postoperative combined intraperitoneal (IP) and intravenous (IV) combination chemotherapy for advanced ovarian carcinoma has been studied in four phase III randomized clinical trials conducted by the GOG. Regional therapy takes advantage of both the prolonged confinement of disease within the peritoneal cavity and the steep dose-response relationship observed for most cytotoxic agents. In addition, by exploiting the peritoneal-plasma barrier, the rate of drug clearance is slowed from the peritoneal to systemic compartment and creates a concentration differential favoring the peritoneal cavity.

Although two previous GOG phase III trials of IP chemotherapy for ovarian carcinoma demonstrated significantly improved OS, they were criticized for not including taxanes (GOG 104) and using dose escalation of IV carboplatin area under the curve (AUC) 9 in the IP arm (GOG 114).^{49,50} For the third GOG study (GOG 0172), Armstrong et al.⁵¹ randomized 415 eligible patients with no residual mass greater than 1 cm after cytoreduction to receive paclitaxel (135 mg/m² IV over 24 hours) followed by either cisplatin (75 mg/m² IV on day 2)—the control IV arm—or cisplatin (100 mg/m² IP on day 2) plus paclitaxel (60 mg/m² IP on day 8)—investigational IP/IV arm.⁵¹ Because paclitaxel is a relatively large molecule, the study team was concerned that without IV paclitaxel, system levels of the taxane would be suboptimal if dependent only on absorption of the drug from the peritoneal cavity. Treatment was repeated every 21 days for 6 cycles. Grades 3 and 4 pain; fatigue; and hematologic, gastrointestinal, metabolic, and neurologic toxic effects were more common in the IP/IV group than in the IV group ($P \leq 0.001$). Only 42% of the patients in the IP/IV arm completed 6 cycles, but the median duration of PFS in the IV arm and IP/IV arm was 18.3 and 23.8 months, respectively ($P = 0.05$). The median duration of OS in the IV and IP/IV arms was 49.7 and 65.6 months, respectively ($P = 0.03$; Fig. 52.2).⁵¹ Quality of life (QOL) was significantly worse in the IP/IV group before cycle 4 and 3 to 6 weeks after treatment but not 1 year after treatment. Although IP/IV therapy significantly improved OS, due to high toxic effects, widespread adoption of IP/IV therapy for advanced ovarian carcinoma has not occurred.⁵¹

GOG protocol 252 was designed to resolve the toxicity issues and validate the survival benefit attributed to IP therapy. Bevacizumab was included as both frontline and maintenance therapy in GOG 252. The three arms tested IP carboplatin, weekly dose-dense paclitaxel, and a reduced intensity and dose-modified GOG-172 regimen. As a result, the trial lacked a definitive control, and the data were not interpretable. Survival data are not yet fully mature but are unlikely to be noteworthy given the negative PFS findings and with the excess toxicity of the IP route, and the ascension of the biologics, there has been a decline in the use of IP chemotherapy.⁵²



No. at Risk					
Intravenous therapy	210	183	157	123	106
Intraperitoneal therapy	205	183	165	142	114

FIGURE 52.2 Overall survival curves demonstrating superiority of intraperitoneal plus intravenous chemotherapy compared with intravenous chemotherapy alone for women with optimally debulked ovarian carcinoma (Gynecologic Oncology Group protocol 0172). (Used with permission. © The Massachusetts Medical Society, 2006. All rights reserved. Armstrong DK, Bundy B, Wenzel L et al. Intraperitoneal cisplatin and paclitaxel in ovarian cancer. *N Engl J Med* 2006;354[1]:34–43.)

Hyperthermic Intraperitoneal Chemotherapy

Hyperthermic intraperitoneal chemotherapy (HIPEC) combines the pharmacokinetic advantage inherent to the intracavitary delivery of certain cytotoxic drugs with the direct cytotoxic effect of hyperthermia. Hyperthermia enhances the tissue penetration of the administered drug. The chemotherapeutic agents used in HIPEC need to have a cell cycle–nonspecific mechanism of action and should ideally show a heat-synergistic cytotoxic effect. Delivery of HIPEC requires an apparatus that heats and circulates the chemotherapeutic solution so that a stable temperature is maintained in the peritoneal cavity during the procedure. HIPEC can be administered by the open Coliseum technique developed by Sugarbaker for the treatment of pseudomyxoma peritonei or by a closed technique, which has the advantage of rapidly achieving and maintaining hyperthermia because there is minimal heat loss. Hyperthermic temperature is monitored using temperature probes placed in the bladder, esophagus, and inflow and outflow IP catheters.

Among patients with platinum-sensitive recurrent ovarian cancer who are candidates for secondary cytoreduction as well as for those with primary advanced disease who have responded to neoadjuvant IV chemotherapy, HIPEC is administered immediately following extensive cytoreductive surgical procedures leaving patients with no gross residual disease (R_0). Perioperative IP chemotherapy regimens that use early postoperative IP chemotherapy (EPIC) use moderate drug dosages for HIPEC, whereas those that do not use EPIC use much higher dosages for HIPEC. In recent years, bidirectional HIPEC regimens (concurrent administration and IV chemotherapy) have gained ground.

In the ovarian cancer setting, HIPEC was first studied in women with recurrent disease. Following optimal secondary cytoreduction for recurrent platinum-sensitive ovarian cancer, Fagotti et al.⁵³ treated 41 patients with oxaliplatin 460 mg/m² heated to 41.5°C for 30 minutes, followed by 6 cycles of IV chemotherapy consisting of oxaliplatin 100 mg/m² plus docetaxel 75 mg/m².⁵³ There were no cases of intraoperative death or death within 30 days of surgery. For 25 patients with a minimum follow-up of 18 months, the median disease-free survival and OS were 24 months (range, 6–60 months) and 38 months (range, 18–60 months), respectively.⁵³

Two recent randomized phase III trials have studied HIPEC in women with newly diagnosed advanced EOC. Van Driel et al.⁵⁴ from the Netherlands conducted a multicenter phase III study to determine whether the addition of IP chemotherapy (cisplatin

100 mg/m²) under hyperthermic conditions to interval cytoreductive surgery would improve the outcome among women receiving neoadjuvant chemotherapy for stage III ovarian cancer. Interval cytoreduction with HIPEC was associated with longer recurrence-free survival than surgery alone (15 vs. 11 months, respectively; HR, 0.65; 95% CI, 0.49 to 0.86; $P = 0.003$). At the time of analysis, 49% of the study population was alive, with a significant improvement in OS favoring HIPEC versus surgery alone (48 vs. 34 months, respectively; HR, 0.64; 95% CI, 0.45 to 0.91; $P = 0.01$). Grade 3 or 4 adverse events were similar in both arms.⁵⁴ The second trial, conducted in Korea and presented by Lim et al.,⁵⁵ included women in whom optimal cytoreductive surgery had been performed either through primary or interval debulking. Randomization to HIPEC (cisplatin 75 mg/m²) or control occurred intraoperatively based on residual tumor <1 cm and was balanced for the use of neoadjuvant chemotherapy and other clinical factors. The 5-year OS was not significantly different between the HIPEC and no HIPEC arms, but for women who received neoadjuvant chemotherapy, the median 5-year OS for the HIPEC arm was 47.9% as compared with 27.7% in the control group.⁵⁵ More follow-up has been scheduled to critically evaluate the impact that HIPEC may have in this subgroup.

Dose-Dense Paclitaxel

Weekly dose-dense administration of paclitaxel has been investigated as a promising therapeutic strategy in the management of primary advanced EOC. The rationale for weekly administration is that more frequent delivery of moderate doses may achieve greater efficacy than standard doses every 3 weeks through more sustained exposure of dividing tumor cells to the cytotoxic effects of paclitaxel, which primarily acts on microtubules to arrest mitosis. The weekly dose-dense regimen was carefully studied by the Japanese Gynecologic Oncology Group (JGOG).

In an effort to harness the antiangiogenic properties of metronomic paclitaxel, on behalf of the JGOG, Katsumata et al.^{56,57} randomized 631 patients with stage II to IV ovarian carcinoma to receive 6 cycles of either paclitaxel (180 mg/m² over 3 hours) plus carboplatin (AUC 6) on day 1 or dose-dense paclitaxel (80 mg/m² over 1 hour) on days 1, 8, and 15 plus carboplatin (AUC 6) on day 1. Cycles were repeated every 21 days, and the primary end point was PFS. Median PFS was longer in the dose-dense treatment group (28.0 months; 95% CI, 22.3 to 35.4 months) than in the conventional treatment group (17.2 months; 95% CI, 15.7 to

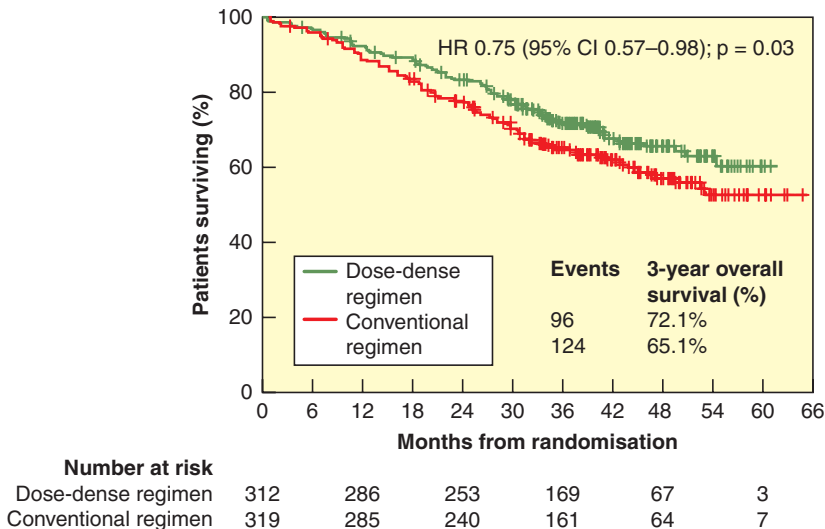


Figure 52.3 Overall survival curves demonstrating superiority of dose-dense weekly paclitaxel (80 mg/m² body surface area [BSA]) versus every-21-day paclitaxel (175 mg/m² BSA) in a phase III randomized trial of Japanese women with advanced ovarian carcinoma. HR, hazard ratio; CI, confidence interval. (Used with permission. © American Society of Clinical Oncology, 2009. All rights reserved. Katsumata N, Yasuda M, Takahashi F, et al. Dose-dense paclitaxel once a week in combination with carboplatin every 3 weeks for advanced ovarian cancer: a phase 3, open-label, randomised controlled trial. *Lancet* 2009;374[9698]:1331–1338.)

21.1 months; HR, 0.71; 95% CI, 0.58 to 0.88; $P = 0.0015$). Neutropenia was the most common adverse event (92% in dose-dense group, 88% in conventional group), and grade 3 or 4 anemia was significantly higher in the dose-dense group than the conventional group, (69% vs. 44%, respectively; $P < 0.0001$).^{56,57} The frequencies of other toxic effects were similar between groups. At a median follow-up of 76.8 months, the investigators reported long-term results, with median PFS and median OS being significantly improved in the dose-dense group (PFS: 28.2 vs. 17.5 months; HR, 0.76; 95% CI, 0.62 to 0.91; $P = 0.0037$; OS: 100.5 vs. 62.2 months; HR, 0.79; 95% CI, 0.63 to 0.99; $P = 0.039$; Fig. 52.3).^{56,57}

The Japanese study provided sufficient data for many in the United States to adopt a weekly dose-dense paclitaxel schedule for newly diagnosed patients. In some centers, both drugs were given weekly as carboplatin (AUC 2) plus paclitaxel (60 mg/m²). The breast cancer literature has suggested that Asian women may metabolize paclitaxel differently than their Western counterparts. Studies in Europe (Multicenter Italian Trial in Ovarian Cancer [MITO]-7, International Collaborative Ovarian Neoplasm [ICON]-8) and in the United States (GOG protocol 262) were conducted to address this consideration.^{58–60} All three were unable to demonstrate a PFS advantage using weekly dose-dense paclitaxel. First, in the Italian study, MITO-7, both carboplatin and paclitaxel were given weekly for 18 weeks, but paclitaxel at 60 mg/m² was not delivered in a dose-dense manner.⁵⁸ The co-primary end points were PFS and QOL. Although weekly scheduling did not result in a PFS advantage, fewer patients on that arm experienced grade 3 to 4 neutropenia, febrile neutropenia, grade 3 or 4 thrombocytopenia, and grade 2 or worse neuropathy. Given that QOL scores were poorer in the standard every-21-day therapy arm, a weekly regimen of carboplatin and paclitaxel may be a reasonable option. In GOG protocol 262, no PFS advantage was demonstrable between weekly and every-21-day paclitaxel in women with newly diagnosed ovarian cancer.⁵⁹ Interestingly, in an unplanned subgroup analysis, the survival curves were relatively indistinguishable for women who received optional bevacizumab (84% of the study population), but when considering the subgroup who did not receive bevacizumab, the curves mirrored what the Japanese investigators reported (PFS, 14.2 vs. 10.3 months for those receiving weekly paclitaxel without bevacizumab versus every-21-day paclitaxel without bevacizumab, respectively; HR, 0.62; 95% CI, 0.40 to 0.95; $P = 0.03$).⁵⁹ This analysis, although not prespecified, suggests that the antiangiogenic properties of paclitaxel that manifest with weekly administration may interfere with vascular endothelial growth factor (VEGF) inhibition conferred through bevacizumab. Finally, ICON-8 recently reported on 1,566 women who were randomized to one of the following three treatment arms: standard every-21-day carboplatin

plus paclitaxel, every-21-day carboplatin (AUC 5 to 6) plus weekly dose-dense paclitaxel (80 mg/m²), or weekly carboplatin (AUC 2) plus weekly dose-dense paclitaxel (80 mg/m²).⁵⁹ Patients who had undergone primary cytoreduction or interval cytoreduction following neoadjuvant chemotherapy were permitted to enroll. Although the investigators demonstrated that dose-dense chemotherapy could be delivered successfully without substantial toxicity, there were no significant improvements in PFS compared with standard dosing and scheduling.⁶⁰ With three negative Western studies, the traction that weekly dose-dense paclitaxel had based on the Japanese study is slowly weakening in centers throughout the United States. The one group in which weekly therapy may still have particular merit is those aged 75+ years as evaluated in the EWOC-1 study using paclitaxel 60 mg/m² and carboplatin AUC 2 D1,8,15 Q28,⁶¹ which is substantially better than single-agent carboplatin in the elderly.

Maintenance Chemotherapy

Up to 70% of patients with advanced ovarian carcinoma who enter into remission after surgery and platinum- and taxane-based chemotherapy will ultimately experience relapse. Because EOC is a chemosensitive disease, it seems reasonable that further exposure to antineoplastic agents after primary therapy would be beneficial. Different strategies have been undertaken to address this concept, including extending initial therapy from 6 to 12 cycles, which is not the same as maintenance therapy, and administration of a non-cross-resistant drug in the remission setting. Failed attempts to prevent recurrence by being pharmacologically proactive during the period of clinical remission have included high-dose chemotherapy with stem cell support, whole abdominal radiotherapy, IP phosphorus-32 (³²P) after negative second-look surgery, and immunotherapy using interferon (IFN)- α and the monoclonal antibody oregovomab. However, significantly improved PFS without increased OS has been associated with maintenance paclitaxel.

After a clinically defined CR at the time of completion of platinum-paclitaxel induction, in GOG protocol 178, Markman et al.⁶² randomized 277 patients to 3 versus 12 doses of monthly paclitaxel (175 mg/m² reduced to 135 mg/m² every 28 days). At interim analysis, a highly significant difference in PFS was observed favoring 12 monthly cycles of consolidation therapy (PFS, 28 vs. 21 months).⁶² The study was stopped by the Data Safety Monitoring Board of the NCI once these interim analysis results became available.⁶³ Subsequent updates of the data have raised the possibility that a subset of patients with low CA 125 levels may enjoy a survival benefit.

GOG protocol 212 was an open-label trial designed to validate the potential PFS benefit of maintenance therapy observed

in GOG 178. In an effort to address the lack of a control arm in GOG 178, the GOG Protocol Development Committee required GOG 212 to have three arms, one of which had no treatment.⁶⁴ After 9 years, during which time 1,157 subjects were enrolled, the NCI's Data and Safety Monitoring Committee voted to release the results of the third scheduled interim analysis, which indicated that the relative death hazards passed the futility boundaries for both taxane regimens (paclitaxel: HR, 1.104; 97.5% CI, 0.884 to 1.38; paclitaxel poliglumex: HR, 0.979; 97.5% CI, 0.781 to 1.23).⁶⁴ Maintenance therapy using either investigational arm was associated with significant gastrointestinal and neurologic adverse events.

MAINTENANCE TARGETED THERAPY

Maintenance Therapy with Bevacizumab

Activity of bevacizumab in ovarian cancer was first demonstrated in three separate phase II studies in the recurrent disease population.^{65–67} Despite nine positive phase III randomized clinical trials involving five antiangiogenesis drugs in newly diagnosed advanced EOC (bevacizumab and nintedanib),^{68–70} maintenance therapy (pazopanib),⁷¹ platinum-sensitive recurrent disease (bevacizumab and cediranib),^{72–74} and platinum-resistant recurrent disease (bevacizumab and trebananib),^{75,76} only bevacizumab has received FDA approval, specifically for treatment of women with newly diagnosed advanced disease, platinum-resistant and platinum-sensitive ovarian cancer (Table 52.5). Finally, based on two provocative randomized phase II trials, chemotherapy-free combinations integrating antiangiogenesis therapy with other targeted therapies are under investigation in phase III trials. These include cediranib and the PARPi, olaparib,⁷⁷ and the combination of bevacizumab with the tumor vascular-disrupting agent (TVDA) fosbretabulin.⁷⁸

GOG 0218

GOG 218 was the first phase III randomized trial of targeted therapy in any gynecologic cancer. Initially, Genentech did not file for bevacizumab use in primary ovarian cancer in the United States because, at the time of public disclosure of the GOG 218 results, the FDA had withdrawn accelerated approval of bevacizumab for breast cancer due to the sponsor's inability to show OS rates and some concerns about toxicity (cardiovascular and gastrointestinal perforation). Originally, when the breast question had been put to the FDA, it had convened an Oncology Drugs Advisory Committee (ODAG), which had voted in December of 2008 five to four against approval, resulting in accelerated approval in February of 2009, at which point the 2-year clock began ticking. On November 18, 2011, the approval for bevacizumab in breast cancer was withdrawn by the FDA.

Although OS was not significantly improved in GOG 0218,⁶⁸ it should be noted that an ad hoc survival analysis among women with stage IV disease was noteworthy, with an OS of 40.6 months in the bevacizumab throughout arm versus 32.8 months in the control arm (HR, 0.72; 95% CI, 0.53 to 0.97). In addition, in both trials, bevacizumab was administered to a predefined maximum number of cycles not until disease progression. This may have limited the long-term therapeutic benefit. Furthermore, post-progression therapy cannot be controlled for, and bevacizumab and multiple other active therapeutic agents were available to participants through multiple lines of subsequent therapy, limiting the ability to demonstrate an impact on survival from the statistical standpoint, even if such an impact existed.

Assessment of adverse events identified only hypertension (grade ≥ 2) as significantly more common in the bevacizumab arms, leading to discontinuation of therapy in 2.4% of subjects. The rates of other adverse events—including gastrointestinal fistula or perforation, proteinuria (grade ≥ 3), neutropenia, thromboembolism,

and wound disruptions—were similar among the three treatment arms.^{68,79} Importantly, gastrointestinal wall disruption (along with bleeding and impaired wound healing) are black boxed warnings in the bevacizumab package insert.

As part of GOG 0218, QOL was evaluated using the Trial Outcome Index (TOI) of the Functional Assessment of Cancer Therapy–Ovary (FACT-O) survey.⁸⁰ Questionnaires were completed before cycles 1, 4, 7, 13, and 22 as well as 6 months after completing the study therapy. The mean FACT-O TOI score increased over the duration of the study, indicating improved QOL. During chemotherapy, quality-of-life scores were slightly lower in the bevacizumab groups than in the control group. However, these differences were lost after completion of chemotherapy.

GOG 0218 led to regulatory approval by the European Medicines Agency (EMA) in 2011 and by the FDA in 2018 for the incorporation of bevacizumab (15 mg/kg every 21 days) with standard platinum- and taxane-based combination chemotherapy (followed by maintenance bevacizumab 15 mg/kg every 21 days for 15 cycles) for primary adjuvant therapy.

MAINTENANCE THERAPY USING PARP INHIBITORS

PARPi(s) were initially approved for treatment and maintenance therapy of women with recurrent ovarian cancer. In the platinum-sensitive setting, three PARPi(s)—olaparib, rucaparib, and niraparib—have been approved by the U.S. FDA among patients who experience a partial or complete response to platinum retreatment. This indication is not biomarker driven. In the platinum-resistant population, olaparib was approved in 2014 for patients with germline *BRCA* mutation who were in need of fourth-line or greater therapy. Rucaparib was approved for patients with either germline or somatic *BRCA* mutation requiring third-line therapy or beyond. Finally, niraparib was approved for patients with homologous recombination deficiency (HRD) tumors who had been previously treated with three or more lines of therapy. Details of the regulatory clinical trials studying these PARPi(s) in the recurrent setting were provided in detail in the previous edition of this book. Beginning in 2018, there are now three approvals for PARPi(s) in the first-line maintenance setting (olaparib, niraparib, and olaparib with bevacizumab). There are preliminary data presented from the OReO study of olaparib and ARIEL4 for rucaparib suggesting that retreatment with PARPi(s) can be effective but with a reduced relative benefit; the earlier use of PARPi(s) in the clinical course of ovarian cancer has become the priority.

Historical Notes

The history of the development of PARPi(s) is reviewed in detail in Chapter 14 and in the references.^{81,82}

CLINICAL IMPLICATIONS OF *BRCA*1/2 MUTATION STATUS

Women carrying a germline *BRCA*1 mutation have an 80% lifetime risk of developing breast cancer and a 50% lifetime risk of developing ovarian cancer.⁸³ Women with a germline *BRCA*2 mutation have a 50% lifetime risk of developing breast cancer and 20% lifetime risk for ovarian cancer.⁸⁴ Approximately 10% to 15% of women with ovarian cancer carry a deleterious germline *BRCA*1/2 mutation. Compared with sporadic disease, ovarian cancer in women with *BRCA*1 mutations is characterized clinically by an earlier age of disease onset (e.g., median age of early to mid-50s), relatively higher response to platinum-based combination chemotherapy, and, among those who achieve clinical remission after primary therapy, longer disease-free intervals prior

TABLE 52.5

Phase III Randomized Trials of Antiangiogenesis Therapy in Ovarian Cancer

Trial	Arms	Grade 3 or 4 AEs	Primary End Point
GOG 218 N = 1,873 Newly diagnosed, incompletely resected stage III and any stage IV; GOG PS 0–2	IV carboplatin (AUC 5) + IV paclitaxel (175 mg/m ²) + placebo followed by maintenance placebo q 3 wk IV carboplatin (AUC 5) + IV paclitaxel (175 mg/m ²) + bevacizumab (15 mg/kg) followed by maintenance placebo q 3 wk IV carboplatin (AUC 5) + IV paclitaxel (175 mg/m ²) + bevacizumab 15 mg/kg followed by maintenance bevacizumab 15 mg/kg q 3 wk	HTN 22.9%; GI events 2.6%; proteinuria 1.6%; VTE 6.7%	Median PFS, 10.3 vs. 11.2 vs. 14.1 mo; HR, 0.717; 95% CI, 0.625–0.824; <i>P</i> < 0.001
ICON-7 N = 1,528 Newly diagnosed, stage I–IIA (clear cell, grade 3); IIB–IV; ECOG PS 0–1	IV carboplatin (AUC 5) + IV paclitaxel (175 mg/m ²) q 3 wk IV carboplatin (AUC 5) + IV paclitaxel (175 mg/m ²) + IV bevacizumab (7.5 mg/kg) + IV bevacizumab (7.5 mg/kg) maintenance q 3 wk	Bleeding 1%; HTN 6%; VTE 4%; GIP 1%; neutropenia 17%	Median PFS, 17.3 vs. 19.0 mo; HR, 0.81; 95% CI, 0.70–0.94; <i>P</i> = 0.0041
AGO-OVAR12/LUME-Ovar 1 N = 1,366 Newly diagnosed, FIGO IIB–IV; ECOG PS 0–2	IV carboplatin (AUC 5–6) + IV paclitaxel (175 mg/m ²) q 3 wk + nintedanib 200 mg PO BID + nintedanib 200 mg PO BID up to 120 wk IV carboplatin (AUC 5–6) + IV paclitaxel (175 mg/m ²) q 3 wk + placebo PO BID + placebo PO BID up to 120 wk	Neutropenia 44%; anemia 14%; thrombocytopenia 18%; diarrhea 20%; elevated ALT 15%; elevated AST 7%; HTN and fatigue 4%	Median PFS, 17.3 vs. 16.6 mo; HR, 0.84; 95% CI, 0.72–0.98; <i>P</i> = 0.0239 (RECIST and CA 125)
AGO-OVAR 16 N = 940 No evidence of progression after surgery and ≥5 cycles of platinum-taxane therapy, FIGO II–IV	Pazopanib 800 mg PO once a day up to 24 mo Placebo PO once a day up to 24 mo	Pazopanib 26% vs. placebo 11%: HTN, diarrhea, nausea, headache, fatigue, neutropenia	Median PFS, 17.9 vs. 12.3 mo; HR, 0.766; 95% CI, 0.64–0.91; <i>P</i> = 0.0021
OCEANS N = 484 Platinum-sensitive recurrence; ECOG PS 0–1	IV carboplatin (AUC 4) + IV gemcitabine (1,000 mg/m ²) + placebo q 3 wk IV carboplatin (AUC 4) + IV gemcitabine (1,000 mg/m ²) + IV bevacizumab (15 mg/kg) q 3 wk	HTN 17.4%; proteinuria 8.5%; bleeding 5.7%; VTE 4%; fistula/abscess 1.6%	Median PFS, 8.4 vs. 12.4 mo; HR, 0.484; 95% CI, 0.388–0.605; <i>P</i> < 0.0001
GOG 213 N = 674 Platinum-sensitive recurrence, after randomization to secondary cytoreductive surgery	IV carboplatin (AUC 5) + IV paclitaxel (175 mg/m ²) + bevacizumab (15 mg/kg) q 3 wk + maintenance bevacizumab (15 mg/kg) q 3 wk IV carboplatin (AUC 5) + IV paclitaxel (175 mg/m ²)	Neutropenia 84%; GIP/abscess 15%; infection 13%; HTN 12%; proteinuria 8%; VTE 4%	Median OS, 42.2 vs. 37.3 mo; HR, 0.827; 95% CI, 0.683–1.005; <i>P</i> = 0.056
ICON-6 N = 456 Platinum-sensitive recurrence, ECOG PS 0–1	Chemotherapy (choice of platinum + paclitaxel; platinum + gemcitabine; carboplatin alone q 3 wk) + PO placebo + continued PO placebo Chemotherapy as above + PO cediranib 20 mg once a day and maintenance PO placebo Chemotherapy as above + PO cediranib 20 mg once a day and maintenance PO cediranib 20 mg	HTN 7%; diarrhea 5%; fatigue 20%; voice change 21%; bleeding 25%	Median PFS, 9.4 vs. 12.5 mo (maintenance vs. chemotherapy-only arm); HR, 0.57; 95% CI, 0.45–0.74; <i>P</i> = 0.024
AURELIA N = 361 Platinum-resistant recurrence; ECOG PS 0–2; no history of SBO; ≤2 prior regimens	IV paclitaxel (80 mg/m ²) days 1, 8, 15, and 22 q 4 wk or IV topotecan (4 mg/m ²) days 1, 8, and 15 q 4 wk or IV pegylated liposomal doxorubicin 40 mg/m ² q 4 wk Chemotherapy as above + IV bevacizumab 10 mg/kg q 2 wk or 15 mg/kg q 3 wk	HTN 20.1%; proteinuria 12.8%; fatigue/asthenia 2.2%; GIP 1.7%; VTE 3.4%	Median PFS, 3.4 vs. 6.7 mo; HR, 0.48; 95% CI, 0.36–0.60; <i>P</i> < 0.00110
TRINOVA-1 N = 919 Recurrent disease (PFI < 12 mo); ≤3 prior lines of chemotherapy; GOG PS 0–1	IV paclitaxel days 1, 8, and 15 q 4 wk + IV placebo weekly IV paclitaxel days 1, 8, and 15 q 4 wk + IV trebananib (15 mg/kg) weekly	Edema 5%; ascites 20%; pleural effusion 13%	Median PFS, 5.4 vs. 7.2 mo; HR, 0.66; 95% CI, 0.57–0.77; <i>P</i> < 0.001

AE, Adverse event; GOG, Gynecologic Oncology Group; IV, intravenous; AUC, area under the curve; q, every; HTN, hypertension; GI, gastrointestinal; VTE, venous thromboembolism; PFS, progression-free survival; HR, hazard ratio; CI, confidence interval; ECOG, Eastern Cooperative Oncology Group; PS, performance status; GIP, gastrointestinal perforation; FIGO, International Federation of Gynecology and Obstetrics; PO, oral; BID, twice a day; ALT, alanine aminotransferase; AST, aspartate aminotransferase; RECIST, Response Evaluation Criteria in Solid Tumors; OS, overall survival; SBO, small bowel obstruction; PFI, progression-free interval.

to recurrence. These clinical features are also present in varying degrees among some women with wild-type *BRCA1/2* genes who exhibit the *BRCA*ness phenotype.

Interestingly, among women with *BRCA1/2* mutations who have undergone RRSO, a disproportionately large number of occult fallopian tube carcinomas (in the absence of a corresponding lesion in the ovary) has been reported. In addition, detailed pathologic study of asymptomatic women with germline *BRCA1/2* mutations has uncovered the likely precursor to high-grade serous carcinoma—*serous tubal intraepithelial carcinomas* (STICs). This pathologic finding is most often identified in secretory epithelial cells in the fimbriated end of the fallopian tube.⁸⁵

Other Genes of Interest

Norquist et al.^{86,87} evaluated 1,915 women with ovarian cancer who had available germline DNA from the University of Washington Gynecologic Tissue Bank ($n = 570$) and from phase III trials of the GOG (GOG 218, $n = 788$; GOG 262, $n = 557$). A total of 208 patients (15%) had *BRCA1/2* mutations, and 8 patients (0.4%) had mutations in DNA mismatch repair genes (e.g., *MSH2*, *MLH1*, *PMS2*, and *MSH6*). Mutations in *BRIP1* ($n = 26$), *RAD51C* ($n = 11$), *RAD51D* ($n = 11$), *PALB2* ($n = 12$), and *BARD1* ($n = 4$) were significantly more common in women with ovarian cancer than in the comparison group composed of the Exome Sequencing Project and Exome Aggregation Consortium. Lilyquist et al.⁸⁸ studied 7,768 ovarian cancer cases of European ancestry with a multigene panel to detect pathogenic alterations. The team identified alterations in 11 genes, with 8 exhibiting a fivefold or greater increased risk for ovarian cancer (*ATM*, *BRCA1*, *BRCA2*, *BRIP1*, *MSH2*, *MSH6*, *RAD51C*, and *RAD51D*). These two studies bring the total number of genes involved in hereditary ovarian cancer to 11.

PARP Inhibitors and Homologous Recombination Deficiency

Due to errors in replication, toxic by-products of cell cycling, and environmental insult, DNA undergoes constant injury, resulting in sequence alterations. Several mechanisms have evolved to facilitate DNA repair, including nucleotide excision repair, base excision repair, homologous recombination, and nonhomologous end-joining.⁸⁹ PARP enzymes are found in the cellular nucleus, where they are activated by DNA damage to identify DNA single-strand breaks. The DNA repair process of single-strand breaks by PARP can be interrupted by the action of PARPi, which interact with binding sites on single-strand breaks in their own zinc finger domains to block PARP and prevent the PARP-mediated transfer of ADP-ribose to form PAR chains.⁸⁹ The summation of these actions by PARPi leads to progression of single-strand breaks to double-strand breaks, which are normally repaired by homologous recombination or repair. Women with germline *BRCA1/2* mutations who develop breast and/or ovarian cancer generally do so in part because of HRD in tissues at risk (ductal and fallopian tube epithelium). In such cases, the complementary *BRCA1/2* allele has been impaired, leading to absence of or defective *BRCA1/2* protein. The loss of high-fidelity homologous recombination in these patients is partly due to an inability to localize DNA polymerase *RAD51*.⁸⁹ In cells with defective homologous repair (e.g., *BRCA1/2* mutations), chromosomal integrity is destroyed, resulting in replication stalling and collapse at the replication fork, cell cycle arrest, and apoptosis.⁸⁹ To summarize, escape strategies to ensure cell survival require intact homologous repair capabilities and/or functional PARP. This is why the provision of a PARPi to *BRCA1/2*-deficient cells is synthetically lethal. Important determinants of the activity of PARPi in ovarian cancer include the molecular signature, platinum sensitivity status, and number of lines of prior therapy.

Genomic Scars

Interestingly, although approximately 15% of women with high-grade serous ovarian cancer have germline *BRCA1/2* mutations, approximately 25% of high-grade serous tumors will have somatic *BRCA1/2* mutations (inclusive of the 15% with germline *BRCA1/2* mutations), and up to 50% will have HRD (inclusive of the 25% with germline and somatic *BRCA1/2* mutations) in general.⁸⁴ The *BRCA1* and *BRCA2* proteins, which facilitate double-strand DNA repair through homologous recombination, are not the only protein products that mediate DNA repair. Examples of genes and gene products intrinsic to pathways that can be targeted by synthetic lethality include the DNA damage response kinases, ataxia-telangiectasia mutated (*ATM*) and ataxia-telangiectasia and Rad3-related (*ATR*), which activate cell-cycle checkpoints, and the key tumor suppressor gene product, *TP53*, a regulator of the cell cycle and apoptotic pathway.⁸⁴ These observations have propelled research in development of an HRD assay that can be validated for clinical use in prospective, randomized trials. HRD testing will identify twice as many patients as somatic testing and 3.5 times as many as germline *BRCA1/2* testing. Although gene sequencing appears to be an attractive method to identify such patients, HRD resulting from epigenetic events such as *BRCA1* promoter hypermethylation will be missed with a pure gene sequencing approach.⁸⁴

Tumors may undergo one or more events that restore high-fidelity homologous recombination, leading to misclassification as having HRD or being sensitive to PARPi. To improve the performance of companion diagnostic HRD assays for PARPi, Watkins et al.⁹⁰ have proposed developing a diagnostic assay integrating a genomic scar-based biomarker as an indicator of PARPi resistance. Conversely, tracing genomic scars back to the specific drivers of the mutator phenotype will allow the selection of therapies to target the specific initiators. In such a context, genomic scars may be viewed as reports of HRD and drug response. HRD leads to at least three manifestations of genomic instability or genomic scarring, including genomic loss of heterozygosity (LOH), telomeric allelic imbalance (TAI), and large-scale state transitions (LST).⁹¹ HRD status may be evaluated by analyzing each of these biomarkers as well as the total number of coding mutations to generate an HRD score. HRD assays would satisfy the FDA's initiative requirement for every new drug to be accompanied to market by a biomarker that predicts its effectiveness.⁸⁹ An example of an FDA-approved HRD assay is myChoice HRD by Myriad Genetics. Table 52.6 contains the initial FDA approvals of PARPi in ovarian cancer.

SOLO-1/GOG-3004 (NCT01844986)

SOLO-1 is an international, 2:1 randomized, double-blind, phase III trial that evaluated the efficacy and tolerability of olaparib tablets (300 mg twice daily) in 391 patients with newly diagnosed advanced epithelial ovarian, tubal, or peritoneal carcinoma harboring germline or somatic *BRCA* mutations. The primary endpoint was PFS. After a median follow-up of 42 months, the risk of disease progression or death was reduced by 70% among the patients receiving olaparib as compared with placebo (HR, 0.30; 95% CI, 0.23 to 0.41; $P < 0.001$).⁹² At that time, the median PFS had not been reached. On December 19, 2018, the FDA approved olaparib tablets as a first-line maintenance therapy for newly diagnosed advanced ovarian cancer for germline or somatic *BRCA* mutation.

In a recent post-hoc analysis containing 5 years of follow-up, the median PFS was 56.0 months with olaparib versus 13.8 months with placebo (HR, 0.33; 95% CI, 0.25 to 0.43).⁹³ No new safety signals were identified.

PAOLA-1 (NCT02477644)

PAOLA-1 is an international, 2:1 randomized, double-blind, international phase trial studying bevacizumab with and without

TABLE 52.6

Initial U.S. Food and Drug Administration Approvals of PARP Inhibitors in Ovarian Cancer

	Niraparib	Olaparib	Rucaparib
Dose Note: *Subsequent modifications	300 mg once daily (*200 mg QD for weight <170 lbs or platelets <150,000/ μ L)	400 mg BID capsules *300 mg BID tablets	600 mg BID
No. of pills daily	3	16 (capsules)	4
No. of prior lines of chemotherapy	≥ 2 (maintenance)	≥ 2 (maintenance) ≥ 4 (treatment)	≥ 2 (maintenance) ≥ 3 (treatment)
Population	Recurrent ovary in complete or partial response to platinum	gBRCA ^{MUT}	gBRCA ^{MUT} or somatic BRCA1/2 mutation
Adverse events	Thrombocytopenia (34%), anemia (25%), neutropenia (20%)	Anemia (18%), abdominal pain (7%), fatigue (6%)	Nausea (25%), anemia (25%), fatigue (11%)

PARP, Poly (ADP-ribose) polymerase; QD, every day; BID, twice a day; gBRCA^{MUT}, germline BRCA1/2 mutation.

All 3 PARPi have the same maintenance indication (recurrent ovary [germline or BRCA wildtype] in complete or partial response to platinum); only rucaparib and olaparib have treatment indications, and only olaparib and niraparib have an upfront switch maintenance indication.

olaparib as a maintenance therapy for 806 patients with newly diagnosed, advanced high-grade ovarian cancer who responded following treatment with first-line platinum and taxane-based chemotherapy plus bevacizumab. The primary end point was investigator-assessed disease progression or death. Using the Myriad's myChoice test with a cutoff of ≥ 42 , 30% of patients had a tBRCA^{MUT} and 48% had HRD. HRD status was unknown in 18% of patients. The study met the primary end point of investigator-assessed PFS in the intention-to-treat population; median PFS was prolonged by 5.5 months with the addition of olaparib maintenance (22.1 months vs. 16.6 months; HR, 0.59; 95% CI, 0.49 to 0.72).⁹⁴ In the BRCA^{MUT} population, the HR was 0.31 (95% CI, 0.20 to 0.47), which was similar to SOLO-1 despite the BRCA^{MUT} patients being overall higher risk and using bevacizumab maintenance as a comparator arm rather than placebo. There was also a significant PFS benefit in the HRD population, including the tBRCA^{WT} HRD-positive cohort. However, without a comparator arm of olaparib alone it is not possible to conclude whether this benefit was from synergy between olaparib and bevacizumab or from olaparib alone. There appeared to be no benefit in the HR-proficient population. The combination of olaparib and bevacizumab was determined to be safe. On May 8, 2020, the FDA expanded the indication of olaparib to include its combination with bevacizumab for first-line maintenance treatment of patients with advanced ovarian cancer who are in complete or partial response to first-line platinum-based chemotherapy and whose cancer is associated with homologous recombination deficiency defined by either a deleterious or suspected deleterious BRCA mutation and/or genomic instability.

In a recent post-hoc analysis of PAOLA-1, the role of mutations involved in homologous recombination repair (HRR), excluding BRCAm, were evaluated as a predictive biomarker for efficacy of maintenance therapy using olaparib plus bevacizumab.

The investigators reported that HRR gene panels beyond somatic BRCAm were unable to select patients who would benefit from this combination and recommend that gene panels should not be substituted for HRD assays based on BRCA and/or genomic instability testing.⁹⁵

PRIMA/ENGOT-OV26/GOG-3012 (NCT02655016)

PRIMA was a trial of niraparib maintenance therapy in patients with newly diagnosed EOC irrespective of BRCA status.⁹⁶ The primary end point was PFS in patients with HRD, followed by those in the overall population using a hierarchical testing strategy. The myChoice HRD test was used to determine HRD status, with the presence of a tumor BRCA^{MUT} (tBRCA^{MUT}, which includes both germline and somatic mutations) or a GIS of ≥ 42 categorizing patients as HRD. Overall, 50.9% of patients were HRD, 34.0% were HR proficient and 15.1% were indeterminant. Thirty percent of patients overall had a tBRCA mutation.

PRIMA was the first study to show that PARP inhibitors given as maintenance therapy in newly diagnosed EOC significantly prolonged PFS regardless of biomarker status. In the overall intent-to-treat population, the risk of progression or death was 38% lower in patients receiving maintenance niraparib compared with placebo (HR, 0.62; 95% CI, 0.50 to 0.76).⁹⁶ In exploratory analysis, the treatment benefit was most pronounced in patients with tBRCA^{MUT} (HR, 0.40; 95% CI, 0.27 to 0.62), followed by BRCA^{WT} HRD tumors (HR, 0.50; 95% CI, 0.30 to 0.83) and HR-proficient tumors (HR, 0.68; 95% CI, 0.49 to 0.94). While PFS is longer in patients with non-BRCA HRD compared with HR proficiency, the hazard ratios in the subgroup analysis overlap considerably, indicating that GIS score alone only weakly discriminates between PARP response. The PFS benefit in patients with HR proficiency also suggests that either PARP has mechanisms of action regardless of HR status or that current HRD tests are suboptimal surrogates for HRD. On April 29, 2020, the FDA approved niraparib (200 mg daily) for primary maintenance therapy of all patients with advanced ovarian cancer who are in complete or partial response to first-line platinum-based chemotherapy.

The patient population in SOLO-1 is quite distinct from PRIMA with respect to risk for disease progression. The SOLO-1 population consisted entirely of patients with deleterious BRCA^{MUT} who are recognized as having a more favorable prognosis, with 82% having a complete response to platinum-based therapy. In contrast, patients enrolled in PRIMA were higher stage (35% were FIGO stage IV), received more neoadjuvant chemotherapy (67%), and only 71% achieved a complete response after platinum-based chemotherapy. Stage III patients were required to have visible residual disease after primary surgery. In subgroup analysis of higher-risk patients in SOLO-1, the hazard ratio for disease progression was 0.49 (95% CI, 0.25 to 0.94) for stage IV disease and 0.44 (95% CI, 0.25 to 0.77) for residual macroscopic disease after debulking surgery, which is similar to the tBRCA^{MUT} subgroup in PRIMA (HR, 0.40; 95% CI, 0.27 to 0.62).

AVANOVA2 (NCT02354131) and OVARIO (NCT03326193)

The efficacy and safety of combining niraparib (300 mg daily) and bevacizumab for treatment of 97 patients with platinum-sensitive ovarian cancer was studied in the open-label, randomized, phase II, superiority trial, AVANOVA2. At a median follow-up of 169 months, the combination significantly improved PFS compared with niraparib alone (median PFS, 11.9 vs. 5.5 months, respectively; HR, 0.35; 95% CI, 0.21 to 0.57; $P < 0.0001$).⁹⁷ The most common grade 3 or worse events in both groups were anemia and

thrombocytopenia; the most common grade 3 or worse event in the combination group was hypertension. There were no treatment-related deaths.

Following up on the results of AVANOVA2, the open-label, single-arm phase II trial, OVARIO, evaluated the combination of niraparib plus bevacizumab as a first-line maintenance therapy in women with newly diagnosed high-grade serous or endometrioid stage IIIB to IV ovarian cancer who had a complete response, partial response, or no evidence of disease after first-line platinum-based chemotherapy plus bevacizumab. At the 18-month landmark analysis, 62% of patients in the overall population, 76% of patients in the HRD subpopulation, and 47% of patients in the HR-proficient subpopulation remained progression free on niraparib plus bevacizumab maintenance.⁹⁸

VELIA/GOG-3005 (NCT02470585)

VELIA addressed the efficacy of concurrent administration of chemotherapy and PARPi followed by PARPi maintenance therapy. The study enrolled 1,140 patients and had three arms: chemotherapy plus veliparib followed by maintenance veliparib (veliparib throughout), chemotherapy plus veliparib followed by placebo maintenance (veliparib-combination only) and chemotherapy plus placebo followed by placebo maintenance (control). All patients underwent gBRCA (BRCAAnalysis) and HRD testing (myChoice HRD). The primary end point of VELIA was PFS in the veliparib-throughout group as compared with the control group, tested hierarchically in the BRCA^{MUT}, HRD, and overall populations. In the veliparib-throughout population, the median PFS for the BRCA^{MUT} cohort was 34.7 months versus 22.0 months for the control arm (HR, 0.44; 95% CI, 0.28 to 0.68; $P < 0.001$).⁹⁹ Improved PFS was also reported in the veliparib-throughout arm for the HRD cohort (HR, 0.57; 95% CI, 0.43 to 0.76) and the overall intent-to-treat population (HR, 0.68; 95% CI, 0.56 to 0.83).

In the veliparib-combination only cohort, there was no PFS benefit observed in any subgroup (BRCA^{MUT}: median PFS 21.1 in veliparib-combination vs. 22.0 months in control; HR, 1.22; 95% CI, 0.82 to 1.80). A significant shortcoming of VELIA is that the study design was modeled after that of GOG-0218 (NCT00262847) in that there was no chemotherapy alone plus maintenance veliparib arm. For this reason, it is not possible to determine whether veliparib provides any benefit when added to chemotherapy and then continued in the maintenance setting.

In contrast to PRIMA, the myChoice GIS cutoff for assigning HRD was ≥ 33 . The lowered threshold increased the sensitivity of the test to PARP response by decreasing false negatives. This resulted in a higher percentage of patients overall classified as HRD (55%) despite having a lower percentage of BRCA^{MUT} patients (26%) than PRIMA.

As the time of this writing, it is uncertain whether there is a regulatory path forward for veliparib for the indication studied in VELIA.

BRCA1/2 REVERSION MUTATIONS

Among women with gBRCAmut, selective pressure through PARPi exposure may induce secondary intragenic (reversion) mutations that may restore BRCA1/2 protein function, leading to clinically significant rates of acquired resistance. Christie et al.¹⁰⁰ obtained plasma and tumor samples from 30 patients with high-grade serous ovarian cancer and either BRCA1 or BRCA2 germline mutations.¹⁰⁰ Five patients were identified with intragenic mutations predicted to restore BRCA1/2 open reading frames. Reversion mutations were detected only in tumor samples from patients with recurrent disease (5 of 16 patients) and only in circulating cell-free DNA from patients with a tumor-detected

reversion (3 of 5 patients). An increasing number of other resistance mechanisms, such as alternate DNA damage repair through the ATR/CHK1 pathway, and secondary somatic mutations that restore RAD51C and/or RAD51D function have been implicated in PARPi resistance.¹⁰¹

TOLERABILITY OF PARP INHIBITORS

PARP appears to be relatively well tolerated, although anemia, nausea, and fatigue are commonly reported. Transaminase elevation has been reported with both rucaparib and veliparib, and elevation in serum creatinine is associated with olaparib and rucaparib. Women undergoing treatment with a PARPi should be counseled regarding gastrointestinal prophylaxis. A proton pump inhibitor can be taken daily, and prescriptions should be filled for antinausea (e.g., 5-hydroxytryptamine-3 [5-HT₃] antagonist) and antidiarrheal medications. Anemia, asthenia, and fatigue can be addressed with packed red cell blood transfusions and good sleep hygiene. If patients are unable to take naps during the day, caffeinated beverages can be used to address fatigue.¹⁰² Niraparib is associated with headache, dizziness, insomnia, anxiety, and hypertension because of being more lipid soluble and having central nervous system penetration.

Myelodysplastic Syndrome and Acute Myeloid Leukemia and Pneumonitis

Separate from the anemia and thrombocytopenia that may manifest as a consequence of myelosuppression, neutropenia may be further exacerbated in the setting of other myelosuppressive anticancer drugs, including DNA-damaging agents. PARPi should not be initiated until patients have recovered from hematologic toxicity caused by prior chemotherapy (i.e., resolution of adverse events to Common Terminology Criteria for Adverse Events [CTCAE] v.4.03 grade 0 or 1). Complete blood count testing at baseline and monthly thereafter while receiving PARPi is indicated. For patients who develop prolonged hematologic toxicities on a PARPi, treatment should be interrupted with weekly blood count monitoring until recovery. If myelotoxicity has not resolved to grade 0 or 1 after 4 weeks, referral to a hematologist is warranted for bone marrow analysis and cytogenetics.

In a single-arm trial of olaparib monotherapy, myelodysplastic syndrome (MDS) or acute myeloid leukemia (AML) was confirmed in 6 of 298 (2%) enrolled patients with germline BRCA1/2-mutated advanced cancers.¹⁰² In a randomized placebo-controlled trial, MDS/AML occurred in 3 of 136 (2%) patients with advanced ovarian cancer treated with olaparib.¹⁰² Across studies, MDS/AML was reported in 22 of 2,618 (<1%) patients treated with olaparib, with the majority of cases ($n = 17$) being fatal. The duration of therapy with olaparib in patients who developed MDS/AML varied from less than 6 months to greater than 2 years. As discussed earlier, 1.3% of patients treated with niraparib in the NOVA study developed MDS/AML (1.2% in the placebo arm). In ARIEL-2, there were no cases of MDS/AML (rucaparib or placebo arms), but two cases of MDS/AML (0.5%) have occurred among 377 ovarian cancer patients treated with rucaparib outside of ARIEL-2. More recent presentations have suggested that the incidence of MDS/AML is substantially higher (8% with olaparib in SOLO2 and 7% with niraparib in NOVA) in patients with BRCA mutations and protracted exposure to a PARPi.

Pneumonitis, including fatal cases, occurred in <1% of patients treated with PARPi across studies.¹⁰² Patients with new or worsening respiratory symptoms (e.g., dyspnea, fever, cough, wheezing) should have treatment interrupted. If pneumonitis is confirmed on diagnostic workup, the PARPi should be discontinued.

MANAGEMENT OF RECURRENT DISEASE

The Concept of Platinum Sensitivity

Management of Patients with Recurrent Disease

The sensitivity of a patient's disease to platinum agents may be categorized as platinum sensitive, partially platinum sensitive, platinum resistant, and platinum refractory. It is important to recognize that the 6-month disease-free interval that separates the platinum-resistant patients from the partially platinum-sensitive patients, as well as the 12-month disease-free demarcation that separates partially platinum-sensitive patients from platinum-sensitive patients, are both arbitrary clinical break points that are not underlined by any known molecular or biologic mechanisms.

Patients who are platinum refractory comprise the highest-risk group because they have not responded to platinum- and taxane-based chemotherapy and often demonstrate evidence of progression during primary treatment with increasing CA 125 levels; worsening ascites, bowel function, and other symptoms; and occasionally metastatic tumor deposits that are enlarging on physical examination and radiologic imaging. Available chemotherapy options are not likely to benefit this group of patients. The most appropriate management plan is early intervention with palliative care services, and patients with a GOG performance status of 0 to 2 should strongly consider screening for participation in a clinical trial evaluating new therapies.

Women who achieve remission after primary therapy and then relapse within 6 months are considered to be platinum resistant and represent a group for whom conventional therapies are probably ineffective or, at best, offer short-term benefit. These patients should also strongly consider participating in clinical trials. In the absence of a new therapeutic study, most patients with platinum-resistant disease are treated with single agents such as pegylated liposomal doxorubicin (PLD) or topotecan and other agents.

Women who experience recurrence between 6 and 12 months after platinum-based first-line chemotherapy are classified as partially platinum sensitive and can benefit from platinum-based reinduction chemotherapy. However, the therapeutic effect is lower than in women characterized as platinum sensitive. Patients described as having partially platinum-sensitive disease may also be considered for participation in clinical trials evaluating new agents and combinations for platinum-sensitive recurrences.

The concept of artificially extending the platinum-free interval in this population with the ultimate goal of reintroducing platinum in the third-line setting has also gained momentum in recent years. This hypothesis is being studied in the phase III randomized International Ovarian Cancer Patients Trial with Yondelis (INO-VATYON) study (PLD plus trabectedin versus PLD plus carboplatin; NCT01379989) and the MITO-8 trial (carboplatin plus paclitaxel followed by PLD versus PLD followed by carboplatin plus paclitaxel with platinum-free interval of 6 to 12 months; NCT00657878).

Patients who experience relapse >12 months after primary platinum-based therapy have platinum-sensitive disease. The therapeutic landscape for these patients is dominated by reinduction of platinum-based chemotherapy. The addition of a second compound to platinum performed favorably against single-agent platinum in the ICON-4 randomized trial.¹⁰³ Carboplatin combined with paclitaxel, gemcitabine, or PLD have emerged as the principle platinum-based chemotherapy doublets. Importantly, the replacement of paclitaxel with PLD in the Caelyx in Platinum-Sensitive Ovarian Cancer (CALYPSO) trial proved the noninferiority of the regimen over carboplatin plus paclitaxel. Interestingly, carboplatin plus PLD was associated with improved PFS and less alopecia, less neuropathy, and less allergic reactions than carboplatin plus paclitaxel.¹⁰⁴

Secondary Cytoreduction: GOG 213 and the DESKTOP Trials

Theoretically, medically and nutritionally optimized patients with a long disease-free interval and solitary isolated abdominal or pelvic recurrences are most likely to enjoy a survival benefit from surgery. In GOG protocol 0213 (NCT00565851), 485 patients with recurrent platinum-sensitive ovarian cancer who had received one previous therapy and had investigator-determined resectable disease (to no macroscopic residual disease) were randomized to undergo secondary surgical cytoreduction and then receive platinum-based chemotherapy or to receive platinum-based chemotherapy alone. Importantly, a second clinically relevant hypothesis that was tested in GOG 213 was whether bevacizumab added to paclitaxel and carboplatin chemotherapy followed by maintenance bevacizumab improves OS. This second hypothesis is discussed separately in this chapter. With respect to the hypothesis concerning secondary cytoreductive surgery, after a median follow-up of 48.1 months, the hazard ratio for death (surgery vs. no surgery) was 1.29 (95% CI, 0.97 to 1.72; $P = 0.08$), which corresponded to a median OS of 50.6 months and 64.7 months, respectively, indicating no benefit of secondary debulking surgery.¹⁰⁵ Furthermore, patient-reported QOL decreased significantly after surgery but did not differ significantly between the two groups after recovery.

Desktop I and Desktop II

The Descriptive Evaluation of Preoperative Selection Criteria for Operability in Recurrent Ovarian Cancer I (DESKTOP I) is a German study that retrospectively evaluated secondary cytoreduction in 267 women from 25 AGO member institutions who experienced disease recurrence. Women who experienced complete resection at secondary debulking had a significantly longer survival compared with women who underwent surgery that left visible residual disease (median OS, 45.2 months vs. 19.7 months, respectively; HR, 3.71; 95% CI, 2.27 to 6.05; $P < 0.001$).¹⁰⁶ The variables associated with complete resection were Eastern Cooperative Oncology Group (ECOG) performance status of 0, FIGO stage at diagnosis, residual tumor after primary surgery, and absence of ascites. These identified criteria were then prospectively evaluated and validated in 129 platinum-sensitive women in the DESKTOP II trial.¹⁰⁷ This was followed by the randomized, controlled, phase III study, DESKTOP III/ENGOT-Ov20, which studied 407 platinum-sensitive patients in first or second relapse.¹⁰⁸ Complete resection was achieved in 67% of the study population, and the median PFS was 14 months without and 19.6 months with secondary cytoreduction (HR, 0.66; 95% CI, 0.52 to 0.83; $P < 0.001$).¹⁰⁸ With the exception of more frequent myelosuppression in the no-surgery arm, there were no significant differences observed in grade ≥ 3 acute adverse events.

Desktop III (NCT01166737)

Although GOG 0213 failed to demonstrate an OS benefit with secondary debulking among women with platinum-sensitive ovarian cancer, the cohort studied was unselected and the analyses may have been confounded by the inclusion of bevacizumab throughout salvage chemotherapy and maintenance thereafter. In DESKTOP III, 407 patients with recurrent platinum-sensitive disease were eligible for randomization to secondary debulking followed by chemotherapy versus chemotherapy alone if they had a positive AGO score (i.e., ECOG PS 0, ascites <500 mL, and complete resection at initial surgery). Complete secondary cytoreductive surgery was successful in 75% of patients and nearly 90% in both arms received a platinum-containing second-line regimen. The median OS was 53.7 with surgery versus 46.2 months without surgery (HR, 0.75; 95% CI, 0.59 to 0.97, $P = 0.03$). Median PFS was

18.4 months versus 14 months (HR, 0.66; 95% CI, 0.54 to 0.82; $P < 0.001$).¹⁰⁹ It is reasonable to offer secondary debulking to women with platinum-sensitive recurrent disease who fulfill DESKTOP III criteria, particularly if PARPi rather than bevacizumab maintenance is used.

Platinum-Based Doublets for Platinum-Sensitive Recurrent Disease

In ICON-4, Parmar et al.¹⁰³ randomized 802 patients with platinum-sensitive ovarian cancer to paclitaxel plus platinum versus conventional platinum-based, physician's choice chemotherapy. At a median follow-up of 42 months, OS and PFS were significantly improved in the paclitaxel plus platinum group (2-year OS, 57% vs. 50%; HR for death, 0.82; 95% CI, 0.69 to 0.97; $P = 0.02$). This multinational phase III study established the paclitaxel-platinum chemotherapy doublet as the standard for platinum-sensitive recurrent disease.¹⁰³

A phase II trial studying paclitaxel (175 mg/m² over a 3-hour infusion) followed by the third-generation platinum analog oxaliplatin (130 mg/m² over a 2-hour infusion) was reported by Viens et al.,¹¹⁰ with results that appear equivalent to other platinum analogs. Brewer et al.¹¹¹ conducted a phase II study of cisplatin (30 mg/m²) plus gemcitabine (600 mg/m²) on days 1 and 8 every 28 days for women with platinum-refractory disease. Among 27 subjects, there were 4 CRs and 5 PRs, giving an overall response of 16%. Thirteen patients (54%) had stable disease. The median time to progression was 5.4 months, and OS time was 14.9+ months. Grade 4 toxicities were primarily hematologic. Pfisterer et al.¹¹² recognized the need to identify a viable nonneurotoxic alternative regimen and randomized 356 patients with platinum-sensitive recurrent ovarian carcinoma to carboplatin alone (AUC 5 on day 1) versus carboplatin (AUC 4 on day 1) plus gemcitabine (1,000 mg/m² on days 1 and 8). Cycles were repeated every 21 days, and the primary objective was PFS. At a median follow-up of 17 months, the HR for PFS favored the carboplatin plus gemcitabine combination (HR, 0.72; 95% CI, 0.58 to 0.90; $P = 0.0031$).¹¹² The doublet was also associated with a significantly higher response rate (47.2% vs. 30.9%, $P = 0.016$). Although myelosuppression was significantly more common in the combination arm, sequelae such as febrile neutropenia or infections were uncommon. Less than 5% of patients on either arm experienced motor or sensory neuropathy. The combination of carboplatin plus gemcitabine was approved in the United States in 2006, which was a surprising ruling because it was based solely on positive PFS results.

CALYPSO was a phase III, randomized, noninferiority trial conducted by Pujade-Lauraine et al.¹⁰⁴ to test the efficacy and safety of carboplatin (AUC 5) plus PLD (30 mg/m²) every 28 days versus carboplatin (AUC 5) plus paclitaxel (175 mg/m²) every 21 days. A minimum of 6 cycles were prescribed, and the patient population comprised 976 women with recurrence >6 months after first- or second-line platinum- and taxane-based therapy. With a median follow-up of 22 months, PFS for the carboplatin-PLD arm (11.3 months) was statistically superior to the PFS in the carboplatin-paclitaxel arm (9.4 months; HR, 0.821; 95% CI, 0.72 to 0.94; $P = 0.005$); median PFS times were 11.3 and 9.4 months, respectively. Although significantly higher rates of hand-foot syndrome, nausea, and mucositis were observed in the carboplatin-PLD arm, overall severe nonhematologic toxicity (36.8% vs. 28.4%, $P < 0.01$) leading to early discontinuation (15% vs. 6%, $P < 0.001$) occurred more frequently in the carboplatin-paclitaxel arm, as did grade ≤ 2 alopecia, hypersensitivity reactions, and sensory neuropathy.¹⁰⁴ CALYPSO constitutes the largest randomized phase III trial in recurrent ovarian cancer. The improved PFS and superior therapeutic index of carboplatin-PLD make this regimen an excellent alternative to conventional platinum-paclitaxel for platinum-sensitive recurrent ovarian carcinoma.

Nonplatinum Chemotherapeutic Options for Platinum-Resistant Recurrent Disease

The registration trial for topotecan was conducted by ten Bokkel Huinink et al.¹¹³ The investigators randomized 226 patients with recurrent disease to paclitaxel (175 mg/m² over 3 hours) or topotecan (1.5 mg/m² 30-minute infusion, days 1 to 5). Cycles were repeated every 21 days. All patients had progressed after one platinum-based regimen, and all had bidimensionally measurable disease. There was a nonsignificant increased response rate among patients treated with topotecan (20.5%) as compared with those treated with paclitaxel (13.2%). Although the median duration of response did not significantly differ, the median time to progression favored topotecan (23 vs. 14 weeks, $P = 0.002$). Neutropenia was significantly more frequent among those treated with topotecan (79% vs. 23%) but was short lasting and noncumulative.¹¹³ Topotecan was approved in 1996 by the FDA and EMA for the treatment of ovarian cancer.

In the PLD registration trial, Gordon et al.¹¹⁴ randomized 474 patients with measurable disease who had experienced recurrence after or had not responded to first-line platinum-based chemotherapy to topotecan (1.5 mg/m² on days 1 to 5 every 21 days) or PLD (50 mg/m² every 28 days).¹¹⁴ Overall response rates, PFS, and median OS were similar between the two arms, but among the platinum-sensitive subset of patients, treatment with PLD was associated with a statistically significant improvement in PFS (28.9 weeks vs. 23.3 weeks, $P = 0.037$) and OS (108 weeks vs. 71.1 weeks, $P = 0.008$).¹¹⁴ PLD was first approved for platinum-refractory ovarian cancer in 1999 and then received full approval for platinum-sensitive recurrent disease in 2005.

In the OVA-301 trial, Monk et al.¹¹⁵ randomized 672 patients with first recurrence to PLD (50 mg/m²) every 21 days versus PLD (30 mg/m²) followed by a 3-hour infusion of trabectedin (1.1 mg/m²) every 21 days. The primary end point was PFS by independent radiology assessment. The combination of trabectedin and PLD, compared with PLD alone, was shown to improve PFS (7.3 months vs. 5.8 months, respectively; HR, 0.79; 95% CI, 0.65 to 0.96; $P = 0.0190$) and overall response rate (27.6% vs. 18.8%, respectively; $P = 0.0080$).¹¹⁵ In addition, an enhanced activity of trabectedin combined with PLD was observed in platinum-sensitive patients, especially in those with a platinum-free interval range of 6 months to 12 months. Based on these results, in 2009, the European Commission granted marketing authorization for trabectedin combined with PLD (a nonplatinum chemotherapy doublet) for the treatment of patients with relapsed platinum-sensitive ovarian cancer.

In a phase II study by the GOG, 51 patients with platinum-resistant recurrent ovarian cancer were treated with pemetrexed (900 mg/m²) IV infusion over 10 minutes every 21 days. In this study, patients were treated until progression or unacceptable toxicity.¹¹⁶ Median PFS was 2.9 months, and median OS was 11.4 months. Although not yet FDA approved for ovarian cancer, based on these results, consensus guidelines recommended pemetrexed as a single agent for platinum-resistant recurrent disease.¹¹⁶

A phase II trial by the GOG evaluated weekly paclitaxel (80 mg/m²) in 48 patients with platinum- and paclitaxel-resistant ovarian cancer.¹¹⁷ An objective response rate of 20.9% was reported, and serious adverse events were relatively uncommon. Finally, Coleman et al.¹¹⁸ reported on the GOG's phase II trial of nanoparticle albumin-bound paclitaxel (Abraxane) administered at 100 mg/m² on days 1, 8, and 15 on a 28-day schedule. Among 47 evaluable patients with platinum-resistant ovarian cancer, 1 CR and 10 PRs were confirmed (23%), with 17 women (36%) having stable disease. The median PFS was 4.5 months, and median OS was 17.4 months. There were no grade 4 events, with the majority of grade 3 events being hematologic ($n = 9$).

EORTC 5595: Early versus Delayed Treatment of Recurrence

Rustin et al.¹¹⁹ randomized 529 patients in an effort to study the perceived benefits of early treatment of recurrent ovarian cancer on the basis of increased CA 125 concentrations as compared with delayed treatment based on clinically evident disease relapse. At the time of randomization, all patients were in complete remission after first-line platinum-based chemotherapy and had a normal CA 125 concentration. Clinical examination and CA 125 measurements were done every 3 months for all patients, with both investigators and patients masked to the CA 125 results. Randomization to early versus delayed treatment took place when the CA 125 level exceeded twice the upper limit of normal levels. Patients assigned to delayed treatment continued to be masked to their CA 125 measurements, with treatment commencing at clinical or symptomatic relapse. The primary end point was OS. At a median follow-up of 56.9 months from randomization and after 370 deaths in the entire study population, there was no evidence of a difference in OS between early and delayed treatment (25.7 months vs. 27.1 months, respectively; HR, 0.98; 95% CI, 0.8 to 1.2; $P = 0.85$), with the earlier use of chemotherapy adding toxicity while not improving survival.¹¹⁹

ANTIANGIOGENESIS THERAPY FOR TREATMENT OF RECURRENT DISEASE

OCEANS

The remaining phase III bevacizumab trials were in platinum-sensitive and platinum-resistant recurrent EOC. The industry-sponsored Ovarian Cancer Study Comparing Efficacy and Safety of Chemotherapy and Antiangiogenic Therapy in Platinum-Sensitive Recurrent Disease (OCEANS) study—a study of carboplatin and gemcitabine plus bevacizumab in patients with ovarian, peritoneal, or fallopian tube carcinoma—evaluated the efficacy and safety of bevacizumab in the treatment of patients with recurrent platinum-sensitive ovarian cancer.⁷² Overall, 484 patients were randomly assigned to carboplatin-gemcitabine plus bevacizumab versus carboplatin-gemcitabine plus placebo for 6 to 10 cycles. Bevacizumab or placebo was then continued until disease progression. Median PFS for the bevacizumab arm was superior to that for the control arm (12.4 months vs. 8.4 months, respectively; HR, 0.484; 95% CI, 0.388 to 0.605; $P < 0.0001$).⁷²

The addition of bevacizumab significantly improved the objective response rate (ORR; 78.5% vs. 57.4% with placebo; $P < 0.0001$), with most responses being PRs. At the time of final PFS analysis, the OS data were immature, with 141 deaths. An additional analysis was conducted at 235 deaths in which the median OS for the placebo arm was 35.2 months and the median OS for the bevacizumab arm was 33.3 months.⁷² These data remain immature. Interestingly, evaluation of treatment after disease progression indicated use of bevacizumab in 31% of subjects on the control arm, potentially confounding OS analysis.

In addition to studying bevacizumab in the platinum-sensitive space, GOG 0213 was also designed to explore the role of secondary cytoreductive surgery. The open-label, randomized phase III trial was conducted throughout the United States, Japan, and South Korea. Eligible patients were those who had achieved a clinical CR to primary platinum-based chemotherapy and had remained disease free for at least 6 months after the last platinum infusion. Using 1:1 randomization, patients were assigned to standard chemotherapy (carboplatin AUC 5 plus paclitaxel 175 mg/m² body surface area [BSA]) every 21 days with and without bevacizumab (15 mg/kg body weight).⁷³ Patients randomized to the bevacizumab arm also received this drug as maintenance therapy every 21 days until disease progression or unacceptable toxicity. Subjects who participated in the surgical objective were randomly

assigned 1:1 to either of the treatment regimens described earlier. The primary end point was OS, analyzed by intent to treat.

Although the analysis for OS was not significant, a sensitivity analysis based on corrected treatment-free interval stratification supports this intervention in women with platinum-sensitive disease. Based on GOG 213 and OCEANS, on December 6, 2016, the FDA approved the combination of bevacizumab with either carboplatin plus paclitaxel or carboplatin plus gemcitabine followed by maintenance bevacizumab for this patient population.

AURELIA

To assess the impact of bevacizumab on oncologic outcome in patients with platinum-resistant recurrent ovarian cancer, Avastin Use in Platinum-Resistant Epithelial Ovarian Cancer (AURELIA)—a study of bevacizumab added to chemotherapy in patients with platinum-resistant ovarian cancer—was initiated.⁷⁵ This randomized, open-label, phase III clinical trial compared chemotherapy (investigator's choice: weekly paclitaxel, weekly topotecan, or PLD) with chemotherapy plus bevacizumab at a dose of 15 mg/kg. Eligible patients had biopsy-proven recurrent epithelial ovarian, primary peritoneal, or fallopian tube carcinoma (disease-free interval of ≤ 6 months), two or fewer prior anticancer regimens, no history of bowel obstruction or abdominal fistula, and no clinical or radiographic evidence of rectosigmoid involvement. Patients were treated to disease progression or unacceptable toxicity, at which point they crossed over to the treatment with chemotherapy alone (bevacizumab arm) or bevacizumab alone (chemotherapy-alone arm).

A total of 361 subjects were enrolled, with 7% having received prior antiangiogenic therapy and 27% having had a disease-free interval of < 3 months. The addition of bevacizumab to chemotherapy nearly doubled the median PFS (6.7 months vs. 3.4 months; HR, 0.48; 95% CI, 0.38 to 0.60; $P < 0.001$), with consistent findings across all subgroups analyzed.⁷⁵ AURELIA is the first randomized phase III trial in women with platinum-resistant disease to meet its primary end point. On November 14, 2014, the FDA approved bevacizumab in combination with paclitaxel, PLD, or topotecan for treatment of patients with platinum-resistant recurrent epithelial ovarian, fallopian tube, or primary peritoneal cancer.

Similar to ICON-7,⁹¹ mature OS data were also presented at the 2013 European Society of Medical Oncology (ESMO) annual meeting, with no significant differences noted between study arms (OS, 16.6 months in bevacizumab arm vs. 13.3 months in chemotherapy-alone arm; HR, 0.85; 95% CI, 0.66 to 1.08; $P = 0.174$).¹²⁰ The study was not powered to detect a statistically significant difference in OS. In addition, post-progression therapy was not monitored in either arm, with additional confounding by the planned post-progression crossover to bevacizumab in the chemotherapy-alone arm. Overall, 72 patients (40%) initially randomized to chemotherapy alone received bevacizumab after documented progression.¹²⁰

Revisiting GOG 218 and ICON-7

Working seemingly backward, Genentech first obtained regulatory approval for bevacizumab in platinum-resistant ovarian cancer in 2014 followed by approval in 2016 for platinum-sensitive disease and for front-line treatment on June 13, 2018.

IMMUNOTHERAPY

Historical Notes

The development of immunotherapy of cancer is reviewed in detail in Chapter 15 and references at the end of this chapter.^{121–133}

Ipilimumab was studied by Hodi et al.¹²⁶ in 11 women with stage IV EOC who had been previously vaccinated with granulocyte-macrophage colony-stimulating factor (GM-CSF) modified

irradiated autologous tumor cells (GVAX). One patient entered into durable remission that lasted over 4 years, and 3 experienced disease stabilization. Tumor response correlated with the CD8⁺ to regulatory T cells (Treg) ratio, indicating potential synergy between anti-cytotoxic T-lymphocyte antigen 4 (anti-CTLA-4) and Treg-depleting therapies. Only two women experienced grade 3 adverse events (gastrointestinal).¹²⁶

Nivolumab was the first anti-programmed cell death protein 1 (PD-1) agent to be studied in EOC. Hamanishi and colleagues reported a phase II trial of 20 women with platinum-resistant EOC treated with at least two prior lines of chemotherapy.^{127,128} Patients received nivolumab at two dose levels (1 or 3 mg/kg). Two CRs were observed in the 3-mg/kg arm, and one PR occurred in the 1-mg/kg arm, providing an ORR of 15%.¹²⁸ Although the partial responder experienced an antitumor response lasting up to 5 months, nivolumab treatment was discontinued after three courses due to adverse events, and the disease recurred 3 months after withdrawal of the patient from the trial. One of the two patients who experienced a CR had an ovarian serous adenocarcinoma, and after two cycles of treatment, her disease remained undetectable for 2 months beyond completion of the 1-year trial of nivolumab. The other complete responder had peritoneal dissemination of a clear cell carcinoma and entered into a durable CR that continued to last beyond the 1 year of study treatment with nivolumab. Grade 3 or 4 adverse events were reported by 8 patients. The most common adverse events were hypothyroidism, lymphocytopenia, fever, transaminitis, rash, fatigue, anemia, arthralgia, and arrhythmia.¹²⁸

At the 2015 ASCO annual meeting, Varga et al.¹²⁹ reported interim results from the ovarian cancer cohort of the phase Ib KEYNOTE-028 trial. Twenty-six patients with recurrent EOC and programmed cell death protein ligand 1 (PD-L1) expression in $\geq 1\%$ of cells in tumor nests or PD-L1-positive bands in stroma were treated with pembrolizumab 10 mg/kg every 2 weeks for up to 2 years.¹²⁹ There was one CR and two PRs, and six patients achieved stable disease. All responses lasted for at least 24 weeks. The best ORR was 11.5%. The most common adverse events reported were fatigue, anemia, and decreased appetite.¹²⁹

To date, one of the largest EOC experiences with anti-PD-L1 therapy was reported at the 2016 ASCO annual meeting by Disis and colleagues (JAVELIN Solid Tumor Phase 1b Trial; NCT01772004).¹³⁰ A total of 124 patients, unselected for PD-L1 expression, were treated with avelumab 10 mg/kg intravenously every 2 weeks in this trial. The median number of prior therapies was four (range, 1–13). Immune-related adverse events occurred in 66.1% of patients, with 6.5% being grade ≥ 3 (increased lipase and elevated creatinine kinase and autoimmune myositis that led to treatment interruption).¹³⁰ Fatigue, nausea, and diarrhea were the most frequent adverse events. There were no treatment-related deaths. The ORR was 9.7% (PR, $n = 12$), and stable disease occurred in 44.4% of patients ($n = 55$). Using a $\geq 1\%$ cutoff for tumor cell staining in 74 patients evaluable for PD-L1 expression, the ORR was 12.3% in PD-L1-positive patients (7 of 57 patients; 95% CI, 5.1% to 23.7%) versus 5.9% in PD-L1-negative patients (1 of 17 patients; 95% CI, 0.1% to 28.7%).¹³⁰ Overall, the median PFS was 11.3 weeks, and median OS was 10.8 months.

RANDOMIZED TRIALS OF IMMUNE CHECKPOINT INHIBITORS

To date, three phase III randomized trials studying checkpoint inhibitors for advanced or recurrent ovarian cancer have been reported. Unfortunately, none of these studies met its primary end point of PFS. In the JAVELIN Ovarian 100 (NCT02718417) and JAVELIN Ovarian 200 (NCT02580058) trials, the anti-PD-L1 avelumab was studied in women with newly diagnosed advanced ovarian cancer and among patients with recurrent disease, respectively. In JAVELIN 100, 998 patients were randomized 1:1:1 to

chemotherapy alone, chemotherapy followed by maintenance avelumab, and chemotherapy plus avelumab followed by maintenance avelumab. The trial was fully enrolled and at interim analysis was terminated due to futility. The stratified HR for PFS was 1.43 with the avelumab maintenance regimen and 1.14 for the avelumab combination regimen.¹³⁴ In JAVELIN 200, 566 patients were randomized 1:1:1 to avelumab alone, avelumab plus pegylated liposomal doxorubicin, or to pegylated liposomal doxorubicin alone. The primary end points were PFS by blinded independent central review (BICR) and OS. Neither end point was significantly improved by avelumab alone or avelumab plus pegylated liposomal doxorubicin.¹³⁵

The IMagyn050 study (NCT03038100) is a multicenter, placebo-controlled, double-blind randomized phase III trial studying the combination of the anti-PD-L1 atezolizumab plus bevacizumab among 1,301 patients with newly diagnosed stage III or IV ovarian cancer who had either undergone primary cytoreductive surgery with macroscopic residual disease or were planned to receive neoadjuvant chemotherapy and interval cytoreduction. Patients were randomly assigned 1:1 to receive atezolizumab or placebo with chemotherapy plus bevacizumab, omitting preoperative bevacizumab in the neoadjuvant cohort. In the intention-to-treat population, the median PFS was 19.5 versus 18.4 months with atezolizumab versus placebo, respectively (HR, 0.92; 95% CI, 0.79 to 1.07; $P = 0.28$).¹³⁶ In the PD-L1-positive population, the median PFS was 20.8 versus 18.5 months, respectively (HR, 0.80; 95% CI, 0.65 to 0.99; $P = 0.038$). There was no significant survival benefit from atezolizumab at interim analysis.

Combinations of PARPi(s) with immune checkpoint inhibitors with or without antiangiogenesis therapy are currently being studied in several phase III randomized trials in patients with newly diagnosed advanced disease. These include, but are not limited to, the evaluation of durvalumab, olaparib, and bevacizumab in DUO-O (NCT03737643), the combination of rucaparib plus nivolumab in ATHENA (NCT03522246), and dostarlimab, niraparib, and bevacizumab in FIRST (NCT03602859). These three trials are anticipated to report in the upcoming 1 to 2 years.

Table 52.7 lists several randomized phase III trials studying checkpoint inhibitors in ovarian cancer that are currently recruiting patients or preparing for imminent activation. JAVELIN Ovarian 200 is expected to be the first of the phase III randomized trials to report in ovarian cancer. Among the novel therapeutic strategies being used, one of the most promising involves combining immune checkpoint inhibition with PARPi in the subgroup of BRCA1/2-mutated women with EOC who have HRD tumors. Because HRD tumors have a high mutational load, they are likely to produce high levels of neoantigens and, therefore, be susceptible to checkpoint inhibitors. In a murine ovarian cancer model, Higuchi et al.¹³¹ demonstrated that CTLA-4 blockade plus PARP inhibition increases survival in mice ($P < 0.0001$). More clinically applicable is the report by Strickland et al.¹³² from the 2015 ASCO annual meeting in which 34 high-grade serous ovarian cancers with germline BRCA1 or BRCA2 mutations were studied alongside 18 tumors without germline or somatic mutations in BRCA1/2 or other homologous recombination genes.¹³² BRCA1/2-mutated tumors exhibited a strong trend for increased CD8⁺ tumor-infiltrating lymphocytes (TILs; $P = 0.05$) compared with non-BRCA1/2-mutated tumors. Overall, there was a significantly higher CD8⁺/CD4⁺ ratio in BRCA1/2-mutated tumors ($P = 0.02$).¹³² Peritumoral CD3⁺ T cells were also significantly increased in the BRCA1/2-mutated cohort ($P = 0.03$). At the 2017 ASCO annual meeting, Friedlander et al.¹³³ reported on a phase I/II dose escalation and expansion trial that combined the anti-PD-1 monoclonal antibody BGB-A317 with the PARPi BGB-0290. Among 38 patients with a variety of solid tumors, 16 had objective responses, including 1 CR in a woman with ovarian cancer. At the 2018 Annual Meeting of the Society of Gynecologic Oncology, two additional studies reported activity with combined PARPi and checkpoint inhibition: TOPACIO, a study of niraparib

TABLE 52.7

Examples of Phase III Randomized Trials of Checkpoint Inhibitors and Novel Combinations in Ovarian Cancer

Trial Name and ClinicalTrials.gov Identifier	Population	Intervention	Primary Objective
IMaGYN050; NCT03038100	Previously untreated EOC	Carboplatin, paclitaxel, and bevacizumab with and without atezolizumab (during and after)	PFS, OS
JAVELIN Ovarian 100 Trial; NCT02718417	Previously untreated EOC	Avelumab in combination with and/or following platinum-based chemotherapy	PFS
ATHENA; NCT03522246	Previously untreated EOC	Rucaparib or nivolumab as single agents or in combination	PFS
JAVELIN Ovarian PARP 100; NCT03642132	Previously untreated EOC	Maintenance talazoparib alone or in combination with avelumab vs. bevacizumab alone	PFS
ATALANTE; NCT02891824	Platinum-sensitive EOC	Bevacizumab ± atezolizumab	PFS
NRG-GY009; NCT02839707	Platinum-resistant EOC	PLD with atezolizumab and/or bevacizumab	OS, PFS, DLT
JAVELIN Ovarian 200 Trial; NCT02580058	Platinum-resistant/refractory EOC	Avelumab ± PLD vs. PLD alone	OS

EOC, Epithelial ovarian cancer; PFS, progression-free survival; OS, overall survival; PLD, pegylated liposomal doxorubicin; DLT, dose-limiting toxicity.

and pembrolizumab¹³⁷ and MEDIOLA, a study of olaparib plus durvalumab.¹³⁸

THERAPEUTIC VACCINES

FANG Maintenance Autologous Tumor Cell Vaccine for Advanced Ovarian Cancer

At the 2015 Society of Gynecologic Oncology Annual Meeting on Women's Cancer Plenary Session, Oh et al.¹³⁹ presented results of a 2:1 randomized phase II study of the FANG vaccine (Gradalis, Carrollton, TX; GM-CSF/bi-shRNAi furin vector-transfected autologous tumor cells). Tumor was harvested at the time of surgical cytoreduction. Patients who achieved a clinical CR after adjuvant chemotherapy were entered to undergo vaccine construction and received 1.0×10^7 cells per intradermal injection monthly for up to 12 doses or were followed per standard of care (i.e., no maintenance therapy).¹³⁹ Patients in the control group were permitted to cross over at the time of progression. Twenty-one patients were randomized (FANG, $n = 14$; observation, $n = 7$).¹³⁹ No significant adverse events were observed. A 92% T-cell activation per interferon (IFN)- γ -enzyme-linked immunospot assay (ELISPOT) response rate was elicited, and there was a marked delay in time to recurrence (7 recurrences at median of 399 days in the FANG group vs. 5 recurrences at a median of 94 days in the no-FANG group).¹³⁹ These provocative results triggered early closure of the study to pursue a phase III trial design involving 382 evaluable patients.

TOLL-LIKE RECEPTORS

VTX-2337 is a novel toll-like receptor 8 (TLR8) agonist that activates human myeloid dendritic cells, monocytes, and natural killer cells to orchestrate the integration of innate and adaptive antitumor responses. VTX-2337 was granted Fast Track designation for ovarian cancer by the FDA on September 2, 2014, with preclinical models suggesting synergy between the TLR8 agonist and PLD.^{140,141} GOG Partners protocol 3003 (NTC01294293) was a randomized phase II study of PLD with and without VTX-2337 in women with recurrent ovarian cancer that was recently reported not to have met either of its co-primary end points (i.e., PFS and OS).¹⁴²

ONCOLYTIC VIRUSES

The oncolytic virus (OV) can be naturally occurring or an engineered virus that is able to exploit cancer-specific changes in cellular signaling.¹⁴³ The effect is not limited to the tumor microenvironment. In fact, the most critical mechanism of OV action appears to be the cytolytic effect on cancer cells that results in antigen release, triggering the cascade of events resulting in the induction of anticancer adaptive immunity.¹⁴³ Adaptive immunity may eradicate distant micrometastases or even oligometastases and provide long-term anticancer immune surveillance. Recently, the FDA approved talimogene laherparepvec, a herpetic OV for treatment of melanoma.¹⁴³ There has been some interest in harnessing the oncolytic properties of the Newcastle disease virus as a treatment for ovarian carcinoma.

CHIMERIC ANTIGEN RECEPTORS

Genetically modified T cells expressing chimeric antigen receptors (CARs) have generated great enthusiasm in acute lymphoblastic leukemia (ALL), where CAR-modified T cells targeting CD19 resulted in complete remission in 27 of 30 (90%) children and adults with refractory disease.^{144,145} Among the patients with durable remissions lasting up to 24 months were 15 patients who had undergone prior stem cell transplantation and two patients with blinatumomab-refractory disease. Although first-generation CARs demonstrated target cell-specific killing in vitro and encouraging preclinical efficacy in murine tumor models, clinical responses of adoptively transferred T cells expressing α -folate receptor-specific CARs in ovarian cancer were not encouraging.¹⁴⁶ Some have speculated that although these first-generation CARs were able to deliver a primary activation signal to T cells (signal 1), the activated T cells were susceptible to anergy or activation-induced cell death in the absence of exogenous costimulation (signal 2).¹⁴⁶ Importantly, although most hematologic tumors express costimulatory molecules, solid tumor cells and antigen-presenting cells in the tumor microenvironment typically lack such molecules.¹⁴⁶ There appears to be tremendous potential with second-generation CARs, and continued improvements in T-cell therapies will refine CAR design even further.¹⁴⁷

On July 11, 2016, due to three potentially treatment-related deaths as a result of cerebral edema, the FDA halted the phase II ROCKET trial designed to study the efficacy and tolerability of the CAR, JCAR015, for patients with adult B-cell ALL.

(NCT02535364). Juno Therapeutics proposed continuing the trial with the omission of fludarabine (which has a known history of eliciting neurotoxicity) and using only cyclophosphamide for preconditioning. The FDA hold was lifted after a few days.

BISPECIFIC T-CELL ENGAGERS

Bispecific antibodies have also garnered enthusiasm due to their versatility, specificity, cost, and ease of production.¹⁴⁸ Bispecific T-cell engager (BiTE) antibodies are a subclass of bispecific antibodies that are specific for CD3 on one arm and a tumor antigen on the other arm. BiTE antibodies can connect any kind of cytotoxic T cell to a cancer cell independent of the traditional requirements for T-cell receptor specificity, costimulation, or peptide antigen presentation.¹⁴⁸ Some have proposed BiTE antibodies as the “missing link” in cancer therapy.¹⁴⁹ Blinatumomab (Blinincyto; Amgen, Thousand Oaks, CA) has emerged as a promising BiTE characterized by its recombinant bscCD19 × CD3 construct and has demonstrated remarkable antitumor activity in a phase II trial of patients with B-cell malignancies (NCT01466179).¹⁵⁰ This has propelled FDA-accelerated approval of the first-in-class blinatumomab antibody for treatment of a rare form of ALL (Philadelphia chromosome-negative ALL) in December 2014 and has also resulted in rapid extension of BiTE technology against a greater repertoire of tumor antigens.

IMMUNE-MEDIATED TOXICITY

The toxicity profiles of different immunologic agents may vary.¹⁵¹ Of particular concern with anti-PD-1 therapy were the three deaths due to autoimmune pneumonitis reported in a phase I trial (discussed earlier). Nishino et al.¹⁵² reported on three patients with melanoma who developed anti-PD-1-associated pneumonitis. Two patients treated with nivolumab developed acute respiratory distress syndrome (ARDS)-pattern pneumonitis, diffuse ground-glass opacities, reticular opacities, consolidation, and traction bronchiectasis that involved all lobes, with decreased lung volumes and effusions. Both patients were admitted to the intensive care unit and received intravenous antibiotic agents, glucocorticoids, and infliximab.¹⁵³ One of the patients improved over the course of 10 weeks, but the second patient died 4 weeks after the diagnosis. The third patient who had also received nivolumab was diagnosed with a nonspecific interstitial pneumonia with ground-glass opacities and reticular opacities in the peripheral and lower lungs.¹⁵² He was treated with oral glucocorticoids as an outpatient and the pneumonitis resolved after 2 weeks, at which point he resumed nivolumab therapy and completed the 2-year treatment period with 12 cycles. He has remained progression free from melanoma, with no recurrent pneumonitis for 39 months at the time of publication.¹⁵² Applications in ovarian cancer will be forthcoming.

Because immune-related adverse events with checkpoint inhibitors may resemble other autoimmune diseases and endocrinopathies, there is a practical need for vigilance on the part of the treating oncologist and the patient. The basic principles of management include periodic holding of therapy and use of corticosteroids. Diagnostic procedures (e.g., colonoscopy in the setting of autoimmune colitis) may be considered but are not mandatory. It should be recognized that discontinuation of therapy due to immune-related adverse events represents the major limitation in the use of immune checkpoint inhibitors.

IMMUNE-RELATED RESPONSE CRITERIA

Due to the recruitment of immune cells and mediators of immune function to the area(s) of primary tumor burden and metastatic foci, conventional response criteria may underestimate the therapeutic benefit of immunotherapy (in particular, checkpoint inhibition) because objective response and sustained disease stabilization may still occur after a perceived or actual increase in tumor burden or appearance of new lesions. In a 327-patient subgroup of advanced melanoma treated with pembrolizumab, Hodi et al.¹⁵³ compared immune-related response criteria (irRC) with RECIST v1.1 and reported that atypical responses were indeed observed in the cohort and that conventional RECIST may underestimate the survival benefit conferred by pembrolizumab in approximately 15% of patients. The investigators speculate that treatment beyond initial progression as defined by RECIST v1.1 may prevent premature cessation of treatment with resultant loss in potential survival gains. In the OAK lung cancer study using the anti-PD-L1 atezolizumab, patients who elected to remain on protocol-directed therapy despite pseudoprogression by RECIST doubled their OS (from approximately 11 to 12 months to 23 to 24 months) compared to those on placebo (median OS, 5 to 6 months).¹⁵⁴ For this reason, OS has traditionally been the corridor through which immunotherapies have been assessed by the FDA. However, with implementation of the Accelerated Approval program, some checkpoint inhibitors have received regulatory approval based on robust ORR. A deep and sustained response may ultimately translate into a significantly improved OS.

PALLIATIVE CARE

The integration of palliative care early adds to quality of life and may improve quantity of time.¹⁵⁵ How depression, nutrition, activity, cortisol, adrenergic drive, and inflammation interact with pathologic physiology is not at all clear, but it is clear that compassion and open dialogue can help during the difficult transition to end-of-life care. For patients with bowel obstruction, use of a G-tube, a gastric drainage tube placed by interventional radiology, can alleviate the distress of vomiting.

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